

Supplementary materials for
“How many labelers do you have?
A closer look at gold-standard labels”

A Experiments on CIFAR-10H dataset

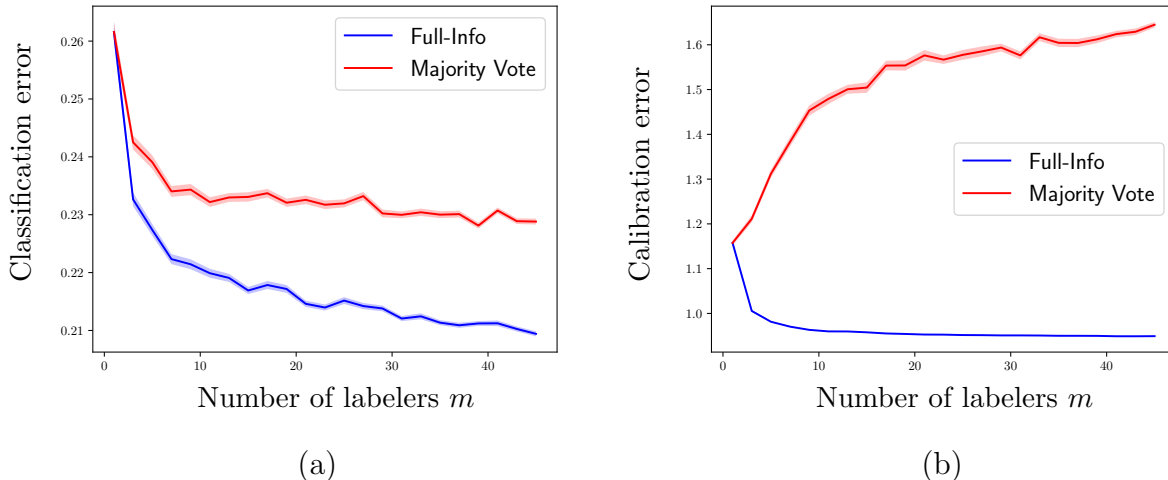


Figure A.1. Experiments on CIFAR-10H dataset. (a) Classification error. (b) Calibration error $|\text{logit}(\hat{p}) - \text{logit}(p)|$ with ResNet features reduced via PCA to dimension $d = 40$. Error bars show 2 standard error confidence bands over $T = 100$ trials.

In this experiment, we consider [Peterson et al.](#)’s [CIFAR-10H](#) dataset [7], which consists of 10,000 images from CIFAR-10 test set with soft labeling in that for each image, we have approximately 50 labels from different annotators. Each 32×32 image in the dataset belongs to one of the ten classes `airplane`, `automobile`, `bird`, `cat`, `dog`, `frog`, `horse`, `ship`, or `truck`; labelers assign each image to one of the classes. To maintain some fidelity to the binary classification setting we analyze throughout the paper, we transform the problem into a set of 10 binary classification problems. For each class c , we take each initial image/label pair $(x, y) \in \mathbb{R}^{32 \times 32} \times \{1, \dots, 10\}$, assigning binary label 1 if $y = c$ and 0 otherwise (so the annotator labels it as an alternative class $y \neq c$). Most of the images in the dataset are very easy to classify: more than 80% have a unanimous label from each of the $m = 50$ labelers, meaning that the MLE and majority vote estimators (2) and (3) coincide for these. (In experiments with this full dataset, we saw little difference between the two estimators.)

As our theoretical results highlight the importance of classifier difficulty, we therefore balance the dataset by considering subsets of harder images as follows. For each fixed target c (e.g., `cat`) and for image i , let $\hat{p}_i$ be the empirical probability of the target among the 50 annotator labels. Then for $p \in [\frac{1}{2}, 1]$, define the subsets

$$\mathcal{S}_p = \{i \in [n] : \max\{\hat{p}_i, 1 - \hat{p}_i\} \leq p\},$$

so that $p = \frac{1}{2}$ corresponds to images with substantial confusion, and $p = 1$ to all images (most of which are easy). We test on $\mathcal{S}_{0.9}$ (labelers have at most 90% agreement), which consists of with 441 images. For image i , we again generate features $x_i \in \mathbb{R}^d$ by taking the last layer of a pretrained ResNet50 neural network $\tilde{x}_i \in \mathbb{R}^{d_{\text{init}}}$, using PCA to reduce to a $d = 40$ -dimensional feature. We follow the same procedure as in Sec. 5.1, subsampling $m = 1, 2, \dots, 45$ labelers and using 10-fold cross validation to evaluate classification and calibration error. Figure A.1 reports results. Again we see that as the number of labelers increases both aggregated and non-aggregated methods evidence improved classification error, but the majority vote procedure (cleaned data) yields less improvement than one with access to all (uncertain) labels. These results are consistent with our theoretical predictions.

B Inconsistency from aggregate labels

To make clear the impossibility of estimation without some knowledge of the link and number m of labelers, we show that there exist distributions with different linear classifiers θ^* and $\bar{\theta}$ generating the same data distribution. Consider a generalized linear model for binary classification, with link $\sigma^* \in \mathcal{F}_{\text{link}}^0$, $\theta^* \in \mathbb{R}^d$, define the model (1) with $\mathbb{P}_{\sigma^*, \theta^*}(Y = y \mid X = x) = \sigma^*(y \langle \theta^*, x \rangle)$, and let $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_d)$, where labels follow $Y_{ij} \stackrel{\text{iid}}{\sim} \mathbb{P}_{\sigma^*, \theta^*}(\cdot \mid X_i)$, $j = 1, \dots, m$; $Y_i^+ = \text{maj}(Y_{i1}, \dots, Y_{im})$ denotes the majority-vote. Then there always exists a

calibration function $\bar{\sigma}$ and label size $\bar{m}$ such that Y_i^+ has identical distribution under both the model (1) and with $\bar{\sigma}, \bar{m}$ replacing σ^* and m . Letting $\mathbb{P}_{(X, Y^+)}^{\sigma^*, \theta^*, m}$ denote the induced distribution on (X_i, Y_i^+) , we have the following formal statement, which we prove in Appendix G.1.

Proposition 8. *Suppose $\sigma^* \in \mathcal{F}_{\text{link}}^0$ satisfies $\sigma^*(t) > 0$ for all $t > 0$. For any $\bar{\theta} \in \mathbb{R}^d$ such that $\bar{\theta} / \|\bar{\theta}\|_2 = \theta^* / \|\theta^*\|_2$ and any positive integer $\bar{m}$, there is another link function $\bar{\sigma}$ such that*

$$\mathbb{P}_{(X, Y^+)}^{\sigma^*, \theta^*, m} \stackrel{\text{dist}}{=} \mathbb{P}_{(X, Y^+)}^{\bar{\sigma}, \bar{\theta}, \bar{m}}.$$

C Technical lemmas

We collect several technical lemmas and their proofs in this section, which will be helpful in the main proofs. See Appendices C.1, C.2, C.3 and C.4 for their proofs.

Lemma C.1. *Suppose $A, B \in \mathbb{R}^{d \times d}$ are symmetric, $AB = BA = 0$ and the matrix $A + B$ is invertible. Then*

$$(A + B)^{-1} = A^\dagger + B^\dagger.$$

The next two lemmas characterize asymptotic behaviors of expectations involving a fixed function f and some random variable Z that satisfies Assumption A2 for given $\beta > 0$ and $c_Z < \infty$. To facilitate stating the theorems, we recall that such Z are (β, c_Z) -regular.

Lemma C.2. *Let $\beta > 0$ and f be a function on $\mathbb{R}_+$ such that $z^{\beta-1}f(z)$ is integrable. If Z is (β, c_Z) -regular (Assumption A2), then*

$$\lim_{t \rightarrow \infty} t^\beta \mathbb{E}[f(t|Z|)] = c_Z \int_0^\infty z^{\beta-1} f(z) dz.$$

Lemma C.3. *Let $\beta > 0$ and $c_Z < \infty$, and Z be (β, c_Z) -regular, let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ satisfy $|f(z)| \leq a_0 + a_p z^p$ for some $a_0, a_p < \infty$ and all $z \in \mathbb{R}$, and assume $|Z|$ has finite p th moment.*

Additionally let $\rho_m(t)$ be the majority vote prediction function (16) and Assumption A4 hold for each σ_j^* with limiting average derivative $\bar{\sigma}^{\star'}(0)$ at zero. Then for any $c > 0$

$$\lim_{m \rightarrow \infty} m^{\frac{\beta}{2}} \mathbb{E} [f(\sqrt{m}|Z|)(1 - \rho_m(cZ))] = c_Z \int_0^\infty z^{\beta-1} f(z) \Phi(-2\bar{\sigma}^{\star'}(0)cz) dz,$$

where $\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$ is the standard normal cumulative distribution function.

The fourth lemma is a uniform convergence result for the empirical risk in the case that we have potentially distinct link functions (cf. Sec. 6).

Lemma C.4. Assume $\mathbb{E}[\|X\|_2^\gamma] \leq M^\gamma$ for some $M \geq 1$ and $\gamma \geq 2$ and let the radius $1 \leq r < \infty$. Let $\mathcal{F}_{\text{link}} \subset \{\sigma : \mathbb{R} \rightarrow [0, 1], \|\sigma\|_{\text{Lip}} \leq \mathbb{L}\}$. Then for a constant $C \lesssim \sqrt{d\mathbb{L}}$, we have

$$\mathbb{E} \left[\sup_{\|\theta\|_2 \leq r} \sup_{\vec{\sigma} \in \mathcal{F}_{\text{link}}^m} |P_n \ell_{\vec{\sigma}, \theta} - L(\theta, \vec{\sigma})| \right] \leq C \cdot (Mr)^{\frac{4\gamma}{3\gamma+1}} \left(\frac{m}{n}\right)^{\frac{\gamma}{3\gamma+1}} \sqrt{\log(rn)}.$$

C.1 Proof of Lemma C.1

As $AB = BA = 0$, the symmetric matrices A and B commute and so are simultaneously orthogonally diagonalizable [4, Thm. 4.5.15]. As $AB = BA = 0$, we can thus write

$$A = U \begin{bmatrix} \Lambda_1 & 0 \\ 0 & 0 \end{bmatrix} U^\top, \quad B = U \begin{bmatrix} 0 & 0 \\ 0 & \Lambda_2 \end{bmatrix} U^\top,$$

for some orthogonal $U \in \mathbb{R}^{d \times d}$, and as $A + B$ is invertible, Λ_1, Λ_2 are invertible diagonal matrices. We conclude the proof by writing

$$(A + B)^{-1} = U \begin{bmatrix} \Lambda_1^{-1} & 0 \\ 0 & \Lambda_2^{-1} \end{bmatrix} U^\top = U \begin{bmatrix} \Lambda_1^{-1} & 0 \\ 0 & 0 \end{bmatrix} U^\top + U \begin{bmatrix} 0 & 0 \\ 0 & \Lambda_2^{-1} \end{bmatrix} U^\top = A^\dagger + B^\dagger.$$

C.2 Proof of Lemma C.2

By the change of variables $w = tz$, we have

$$t^\beta \mathbb{E}[f(t|Z|)] = \int_0^\infty f(tz) \cdot t^{\beta-1} p(z) \cdot t dz = \int_0^\infty f(w) \cdot t^{\beta-1} p(w/t) dw$$

$$= \int_0^\infty w^{\beta-1} f(w) \cdot (w/t)^{1-\beta} p(w/t) dw.$$

As $|w^{\beta-1} f(w) \cdot (w/t)^{1-\beta} p(w/t)| \leq \sup_{z \in (0, \infty)} z^{1-\beta} p(z) \cdot |w^{\beta-1} f(w)|$, where $w^{\beta-1} f(w)$ is integrable by assumption, we can invoke dominated convergence to see that

$$\lim_{t \rightarrow \infty} t^\beta \mathbb{E}[f(t|Z|)] = \int_0^\infty w^{\beta-1} f(w) \cdot \lim_{t \rightarrow \infty} (w/t)^{1-\beta} p(w/t) dw = c_Z \int_0^\infty w^{\beta-1} f(w) dw.$$

C.3 Proof of Lemma C.3

By rescaling arguments, it suffices to prove the theorem for any function f satisfying $|f(z)| \leq 1 + z^p$ on $\mathbb{R}_+$ for any $p \in \mathbb{N}$ such that $|Z|$ has finite p th moment. For such f , we wish to show

$$\lim_{m \rightarrow \infty} m^{\frac{\beta}{2}} \mathbb{E} [f(\sqrt{m}|Z|)(1 - \rho_m(cZ))] = c_Z \int_0^\infty z^{\beta-1} f(z) \Phi(-2\bar{\sigma}^{\star'}(0)cz) dz,$$

where Φ is the standard normal cdf. The key insight is that we can approximate $1 - \rho_m(t)$ by a suitable Gaussian cumulative distribution function (recognizing that $\rho_m(t) > \frac{1}{2}$ for $t \neq 0$ by definition (16) as the probability the majority vote is correct given margin $\langle \theta^\star, X \rangle = t$).

We first assume $Z \geq 0$ with probability 1, as the general result follows by writing $Z = (Z)_+ - (-Z)_+$. We decompose into two expectations, depending on Z being large or small:

$$\begin{aligned} & m^{\frac{\beta}{2}} \mathbb{E} [f(\sqrt{m}|Z|)(1 - \rho_m(cZ))] \\ &= \underbrace{m^{\frac{\beta}{2}} \mathbb{E} \left[f(\sqrt{m}Z)(1 - \rho_m(cZ)) \mathbb{1} \left\{ 0 \leq Z \leq \frac{M}{\sqrt{m}} \right\} \right]}_{\text{(I)}} + \underbrace{m^{\frac{\beta}{2}} \mathbb{E} \left[f(\sqrt{m}Z)(1 - \rho_m(cZ)) \mathbb{1} \left\{ Z > \frac{M}{\sqrt{m}} \right\} \right]}_{\text{(II)}}. \end{aligned} \tag{1}$$

The proof consists of three main parts.

1. We approximate $1 - \rho_m(t)$ by a Gaussian cdf.

2. We can approximate term (I) by replacing $1 - \rho_m(t)$ with the Gaussian cdf, showing that

$$\left| \lim_{m \rightarrow \infty} (\text{I}) - c_Z \int_0^\infty z^{\beta-1} f(z) \Phi(-2\bar{\sigma}^*(0)cz) dz \right| = o_M(1). \quad (2)$$

3. For term (II), we show $1 - \rho_m(cz)$ is small when $Z > M/\sqrt{m}$, which allows us to show that

$$\limsup_{m \rightarrow \infty} |(\text{II})| = o_M(1).$$

Thus by adding the two preceding displays and taking $M \rightarrow \infty$, we obtain the lemma.

Before we dive into further details, we use the shorthand functions

$$\sigma_m^{\text{std}}(z) := \frac{\sum_{j=1}^m (\sigma_j^*(cz) - \frac{1}{2})}{\sqrt{\sum_{j=1}^m \sigma_j^*(cz)(1 - \sigma_j^*(cz))}}, \quad \Delta_m(z) := 1 - \rho_m(cz) - \Phi(-\sigma_m^{\text{std}}(z)), \quad (3)$$

and we also write $p_\infty(\beta) := \sup_{z \in (0, \infty)} z^{1-\beta} p(z) < \infty$.

Part 1. Normal approximation for $1 - \rho_m(t)$ when $t = O(1/\sqrt{m})$. Let $p_j = \sigma_j^*(t)$ for shorthand and $Y_j \sim \text{Bernoulli}(p_j)$ be independent random variables. For $t > 0$, then

$$1 - \rho_m(t) = \mathbb{P}\left(Y_1 + \dots + Y_m < \frac{1}{2}\right) = \mathbb{P}\left(\frac{\sum_{j=1}^m (Y_j - p_j)}{\sqrt{\sum_{j=1}^m p_j(1 - p_j)}} < -\frac{\sum_{j=1}^m (p_j - \frac{1}{2})}{\sqrt{\sum_{j=1}^m p_j(1 - p_j)}}\right).$$

Consider the centered and standardized random variables $\xi_j = (Y_j - p_j)/\sqrt{\sum_{j=1}^m p_j(1 - p_j)}$ so that $\xi_1, \dots, \xi_m$ are zero mean, mutually independent, and satisfy

$$\begin{aligned} \sum_{j=1}^m \text{Var}(\xi_j) &= 1, \\ \sum_{j=1}^m \mathbb{E}[|\xi_j|^3] &= \frac{\sum_{j=1}^m p_j(1 - p_j)(p_j^2 + (1 - p_j)^2)}{(\sum_{j=1}^m p_j(1 - p_j))^{3/2}} \leq \frac{\max_{1 \leq j \leq m} (p_j^2 + (1 - p_j)^2)}{\sqrt{\sum_{j=1}^m p_j(1 - p_j)}}. \end{aligned}$$

By the Berry-Esseen theorem (cf. Chen et al. [1], Shevtsova [10]), for all $t > 0$

$$\left| 1 - \rho_m(t) - \Phi\left(-\frac{\sum_{j=1}^m (p_j - \frac{1}{2})}{\sqrt{\sum_{j=1}^m p_j(1 - p_j)}}\right) \right| \leq \frac{3}{4} \cdot \frac{\max_{1 \leq j \leq m} (p_j^2 + (1 - p_j)^2)}{\sqrt{\sum_{j=1}^m p_j(1 - p_j)}}.$$

Fix any $M < \infty$. Then for $0 \leq t \leq cM/\sqrt{m}$ and large enough m , the right hand side of the preceding display has the upper bound $2/\sqrt{m}$, as in numerator we have $p_j^2 + (1 - p_j)^2 \leq 1$, and in denominator $\min_{1 \leq j \leq m} p_j(1 - p_j) \rightarrow 1/4$ for all j as $m \rightarrow \infty$, which follows from Assumption A4 that

$$\frac{1}{2} \leq \limsup_{m \rightarrow \infty} \max_{1 \leq j \leq m} p_j \leq \limsup_{m \rightarrow \infty} \sup_{1 \leq j < \infty} \sigma_j^* \left(\frac{cM}{\sqrt{m}} \right) = \frac{1}{2}.$$

By repeating the same argument for $-cM/\sqrt{m} \leq t < 0$, we obtain that for large m and $|t| \leq cM/\sqrt{m}$,

$$\left| 1 - \rho_m(t) - \Phi \left(- \left| \frac{\sum_{j=1}^m (\sigma_j^*(t) - \frac{1}{2})}{\sqrt{\sum_{j=1}^m \sigma_j^*(t) (1 - \sigma_j^*(t))}} \right| \right) \right| \leq \frac{2}{\sqrt{m}}. \quad (4)$$

Part 2. Approximating (I) by Gaussian cdf. For the first term (I) in (1), we further decompose into a normal approximation term and an error term,

$$(I) = \underbrace{m^{\frac{\beta}{2}} \mathbb{E} \left[f(\sqrt{m}Z) \Phi(-\sigma_m^{\text{std}}(z)) \mathbf{1} \left\{ 0 \leq Z \leq \frac{M}{\sqrt{m}} \right\} \right]}_{\text{(III)}} + \underbrace{m^{\frac{\beta}{2}} \mathbb{E} \left[f(\sqrt{m}Z) \Delta_m(z) \mathbf{1} \left\{ 0 \leq Z \leq \frac{M}{\sqrt{m}} \right\} \right]}_{\text{(IV)}},$$

where $\Delta_m(z) = 1 - \rho_m(cz) - \Phi(-\sigma_m^{\text{std}}(z))$ as in def. (3). We will show (IV) $\rightarrow 0$ and so (III) dominates. By the change of variables $w = \sqrt{m}z$, we can further write (III) as

$$\begin{aligned} \text{(III)} &= m^{\frac{\beta}{2}} \int_0^{\frac{M}{\sqrt{m}}} f(\sqrt{m}z) \Phi(-\sigma_m^{\text{std}}(z)) \cdot p(z) dz \\ &= \int_0^M w^{\beta-1} f(w) \Phi \left(-\sigma_m^{\text{std}} \left(\frac{w}{\sqrt{m}} \right) \right) \cdot (w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) dw. \end{aligned}$$

We want to take the limit $m \rightarrow \infty$ and apply dominated convergence theorem. Because $(w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) \leq p_\infty(\beta) < \infty$ and $\sigma_m^{\text{std}}(w/\sqrt{m}) \geq 0$, we have

$$w^{\beta-1} f(w) \Phi \left(-\sigma_m^{\text{std}} \left(\frac{w}{\sqrt{m}} \right) \right) \cdot \left(-\sigma_m^{\text{std}} \left(\frac{w}{\sqrt{m}} \right) \right) \cdot (w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) \leq w^{\beta-1} f(w) \cdot \Phi(0) p_\infty(\beta).$$

As $\beta > 0$ and $|f(w)| \leq 1 + w^p$, $w^{\beta-1}f(w)$ is integrable on $[0, M]$, and by (15a) and (15c) in Assumption A4,

$$\lim_{m \rightarrow \infty} \sigma_m^{\text{std}} \left(\frac{w}{\sqrt{m}} \right) = \lim_{m \rightarrow \infty} \frac{\sqrt{m} \left(\bar{\sigma}_m^* \left(\frac{cw}{\sqrt{m}} \right) - \frac{1}{2} \right)}{\sqrt{\frac{1}{m} \sum_{j=1}^m \sigma_j^* \left(\frac{cw}{\sqrt{m}} \right) \left(1 - \sigma_j^* \left(\frac{cw}{\sqrt{m}} \right) \right)}} = 2\bar{\sigma}^{*'}(0)cw.$$

Using the above display and that $\lim_{m \rightarrow \infty} (w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) = c_Z$, we can thus apply dominated convergence theorem to conclude that

$$\lim_{m \rightarrow \infty} \text{(III)} = c_Z \int_0^M w^{\beta-1} f(w) \cdot \Phi(-2\bar{\sigma}^{*'}(0)cw) dw = c_Z \int_0^\infty w^{\beta-1} f(w) \Phi(-2\bar{\sigma}^{*'}(0)cw) dw + o_M(1).$$

Next we turn to the error term (IV). By the bound (4), $|\Delta_m(z)| \leq 2/\sqrt{m}$ when $|z| \leq M/\sqrt{m}$ for large enough m , and substituting $w = \sqrt{m}z$,

$$\begin{aligned} |(\text{IV})| &\leq m^{\frac{\beta}{2}} \mathbb{E} \left[|f(\sqrt{m}Z)| \cdot \frac{2}{\sqrt{m}} \cdot \mathbb{1} \left\{ 0 \leq Z \leq \frac{M}{\sqrt{m}} \right\} \right] \\ &= \frac{2}{\sqrt{m}} \int_0^{\frac{M}{\sqrt{m}}} \sqrt{m} \cdot |f(\sqrt{m}z)| \cdot m^{\frac{\beta-1}{2}} p(z) dz = \frac{2}{\sqrt{m}} \int_0^M w^{\beta-1} |f(w)| \cdot (w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) dw. \end{aligned}$$

By using Assumption A2 again that $(w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) \leq p_\infty(\beta) < \infty$, we further have

$$|(\text{IV})| \leq \frac{2p_\infty(\beta)}{\sqrt{m}} \cdot \int_0^M w^{\beta-1} |f(w)| dw \leq \frac{2p_\infty(\beta)}{\sqrt{m}} \cdot \left(\frac{M^\beta}{\beta} + \frac{M^{p+\beta}}{p+\beta} \right) \rightarrow 0,$$

where we use $|f(w)| \leq 1 + w^p$. We have thus shown the limit (2).

Part 3. Upper bounding (II). In term (II), when Z is large, the key is that the quantity

$1 - \rho_m(t)$ is small when $|t| \geq cM/\sqrt{m}$: Hoeffding's inequality implies the tail bound

$$0 \leq 1 - \rho_m(t) \leq e^{-2(\bar{\sigma}_m^*(t) - \frac{1}{2})^2 m}.$$

Thus

$$|(\text{II})| \leq \int_{\frac{M}{\sqrt{m}}}^1 \sqrt{m} |f(\sqrt{m}z)| e^{-2(\bar{\sigma}_m^*(cz) - \frac{1}{2})^2 m} \cdot m^{\frac{\beta-1}{2}} p(z) dz + \int_1^\infty m^{\frac{\beta}{2}} |f(\sqrt{m}z)| e^{-2(\bar{\sigma}_m^*(cz) - \frac{1}{2})^2 m} \cdot p(z) dz.$$

Using the assumption that $\gamma := \liminf_m \inf_{t \geq c} (\bar{\sigma}_m^*(t) - \frac{1}{2}) > 0$ from (15b), that $|f(z)| \leq 1 + z^p$ and $|Z|$ has finite p th moment, we observe that

$$\int_1^\infty m^{\frac{\beta}{2}} |f(\sqrt{m}z)| e^{-2(\bar{\sigma}_m^*(cz) - \frac{1}{2})^2 m} \cdot p(z) dz \leq m^{\frac{1}{2}(\beta+p)} e^{-2\gamma^2 m} \int_1^\infty (1+z^p) p(z) dz \rightarrow 0,$$

and consequently

$$\begin{aligned} \limsup_{m \rightarrow \infty} |(\text{II})| &\leq \limsup_{m \rightarrow \infty} \int_{\frac{M}{\sqrt{m}}}^1 \sqrt{m} |f(\sqrt{m}z)| e^{-2(\bar{\sigma}_m^*(cz) - \frac{1}{2})^2 m} \cdot m^{\frac{\beta-1}{2}} p(z) dz \\ &= \limsup_{m \rightarrow \infty} \int_M^{\sqrt{m}} w^{\beta-1} |f(w)| e^{-2(\bar{\sigma}_m^*(\frac{cw}{\sqrt{m}}) - \frac{1}{2})^2 m} \cdot (w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) dw. \end{aligned}$$

For $w \in [M, \sqrt{m}]$, we have

$$\left(\bar{\sigma}_m^* \left(\frac{cw}{\sqrt{m}} \right) - \frac{1}{2} \right)^2 m = \left(\frac{\bar{\sigma}_m^* \left(\frac{cw}{\sqrt{m}} \right) - \frac{1}{2}}{\frac{cw}{\sqrt{m}}} \right)^2 c^2 w^2 \geq \left(\inf_{0 < t \leq c} \frac{\bar{\sigma}_m^*(t) - \frac{1}{2}}{t} \right)^2 c^2 w^2,$$

while Assumption (15b) gives $\delta := \inf_{0 < t \leq c} \frac{\bar{\sigma}_m^*(t) - \frac{1}{2}}{t} > 0$, so

$$\limsup_{m \rightarrow \infty} |(\text{II})| \leq \int_M^{\sqrt{m}} w^{\beta-1} |f(w)| e^{-\delta w^2} \cdot (w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) dw.$$

Using the inequality $(w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) \leq p_\infty(\beta) < \infty$, we apply dominated convergence:

$$\lim_{m \rightarrow \infty} \int_M^{\sqrt{m}} w^{\beta-1} |f(w)| e^{-\delta w^2} \cdot (w/\sqrt{m})^{1-\beta} p(w/\sqrt{m}) dw = c_Z \int_M^\infty w^{\beta-1} |f(w)| e^{-\delta w^2} dw = o_M(1).$$

C.4 Proof of Lemma C.4

We follow a typical symmetrization approach, then construct a covering that we use to prove the lemma. Let $P_n^0 = n^{-1} \sum_{i=1}^n \varepsilon_i 1_{X_i, Y_i}$ be the (random) symmetrized measure with point masses at (X_i, Y_i) for $Y_i = (Y_{i1}, \dots, Y_{im})$. Then by a standard symmetrization argument, we have

$$\mathbb{E} \left[\sup_{\|\theta\|_2 \leq r} \sup_{\vec{\sigma} \in \mathcal{F}_{\text{link}}^m} |P_n \ell_{\vec{\sigma}, \theta} - L(\theta, \vec{\sigma})| \right] \leq 2 \mathbb{E} \left[\sup_{\|\theta\|_2 \leq r} \sup_{\vec{\sigma} \in \mathcal{F}_{\text{link}}^m} |P_n^0 \ell_{\vec{\sigma}, \theta}| \right]. \quad (5)$$

We use a covering argument to bound the symmetrized expectation (5). Let $R < \infty$ to be chosen, and for an (again, to be determined) $\epsilon > 0$ let $\mathcal{G} \subset \mathcal{F}_{\text{link}}$ denote an ϵ -cover of $\mathcal{F}_{\text{link}}$ in the supremum norm on $[-R, R]$, that is, $\|g - \sigma\| = \sup_{t \in [-R, R]} |g(t) - \sigma(t)|$, and so for each $\sigma \in \mathcal{F}_{\text{link}}$ there exists $g \in \mathcal{G}$ such that $\|g - \sigma\| \leq \epsilon$. Then [12, Ch. 2.7] we have $\log \text{card}(\mathcal{G}) \leq O(1) \frac{RL}{\epsilon}$. Let Θ_ϵ be a minimal ϵ -cover of $\{\theta \mid \|\theta\|_2 \leq r\}$ in $\|\cdot\|_2$, so that $\log \text{card}(\Theta_\epsilon) \leq d \log(1 + \frac{2r}{\epsilon})$ and $\max_{\theta \in \Theta_\epsilon} \|\theta\|_2 \leq r$. We claim that for each $\|\theta\|_2 \leq r$ and $\sigma \in \mathcal{F}_{\text{link}}$, there exists $v \in \Theta_\epsilon$ and $g \in \mathcal{G}$ such that

$$\left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (\ell_{\sigma, \theta}(Y_{ij} \mid X_i) - \ell_{g, v}(Y_{ij} \mid X_i)) \right| \leq \epsilon + \frac{1}{n} \sum_{i=1}^n \|X_i\|_2 \epsilon + \frac{1}{n} \sum_{i=1}^n 1\{\|X_i\|_2 \geq R/r\}. \quad (6)$$

Indeed, for any $g \in \mathcal{F}_{\text{link}}$ and $\theta, v \in \mathbb{R}^d$, we have

$$\begin{aligned} |\ell_{\sigma, \theta}(y \mid x) - \ell_{g, v}(y \mid x)| &= \left| \int_0^{y \langle x, \theta \rangle} \sigma(-t) dt - \int_0^{y \langle x, v \rangle} g(-t) dt \right| \\ &\leq \sup_{|t| \leq r \|x\|_2} |\sigma(t) - g(t)| + \left| \int_{y \langle x, v \rangle}^{y \langle x, \theta \rangle} |\sigma(-t) - g(-t)| dt \right| \\ &\leq \|\sigma - g\| + 1\{\|x\|_2 \geq R/r\} + |\langle x, \theta - v \rangle| \\ &\leq \|\sigma - g\| + \|x\|_2 \|\theta - v\|_2 + 1\{\|x\|_2 \geq R/r\}, \end{aligned}$$

where we have used that $\sigma, g \in [0, 1]$. Taking the elements g, v in the respective coverings to minimize the above bound gives the guarantee (6).

We now leverage inequality (6) in the symmetrization step (5). We have

$$\begin{aligned} &\mathbb{E} \left[\sup_{\|\theta\|_2 \leq r} \sup_{\vec{\sigma} \in \mathcal{F}_{\text{link}}^m} |P_n \ell_{\vec{\sigma}, \theta} - L(\theta, \vec{\sigma})| \right] \\ &\lesssim \mathbb{E} \left[\max_{\theta \in \Theta_\epsilon} \max_{\vec{g} \in \mathcal{G}^m} |P_n^0 \ell_{\vec{g}, \theta}| \right] + \epsilon + \mathbb{E}[\|X_1\|_2] \epsilon + \frac{1}{n} \sum_{i=1}^n \mathbb{P}(\|X_i\|_2 \geq R/r) \\ &\stackrel{(i)}{\lesssim} \sqrt{\frac{dmRL}{\epsilon} \log \left(1 + \frac{2r}{\epsilon} \right)} \mathbb{E} \left[\frac{r^2}{n} \sum_{i=1}^n \|X_i\|_2^2 \right]^{1/2} + \epsilon + \mathbb{E}[\|X_1\|_2] \epsilon + \frac{\mathbb{E}[\|X_i\|_2^\gamma]}{(R/r)^\gamma} \\ &\leq Mr \sqrt{\frac{dmRL}{n\epsilon} \log \left(1 + \frac{2r}{\epsilon} \right)} + (M+1)\epsilon + \frac{M^\gamma}{(R/r)^\gamma}, \end{aligned}$$

where inequality (i) uses that if Z_i are τ^2 -sub-Gaussian, then $\mathbb{E}[\max_{i \leq N} |Z_i|] \leq \sqrt{2\tau^2 \log N}$, and that conditional on $\{X_i, Y_i\}_{i=1}^n$, the symmetrized sum $\sum_{i=1}^n \varepsilon_i \frac{1}{m} \sum_{j=1}^m \ell_{g_j, \theta}(Y_{ij} | X_i)$ is $r^2 \sum_{i=1}^n \|X_i\|_2^2$ -sub-Gaussian, as $|\ell_{g_j, \theta}(Y_{ij} | X_i)| \leq |\langle X_i, \theta \rangle| \leq r \|X_i\|_2$. We optimize this bound to get the final guarantee of the lemma: set $\epsilon = (Rm/n)^{1/3}$ (and note that we will choose $R \geq 1$) to obtain

$$\mathbb{E} \left[\sup_{\|\theta\|_2 \leq r} \sup_{\vec{\sigma} \in \mathcal{F}_{\text{link}}^m} |P_n \ell_{\vec{\sigma}, \theta} - L(\theta, \vec{\sigma})| \right] \lesssim \sqrt{d \mathbb{L} \log(rn)} M r \left(\frac{Rm}{n} \right)^{1/3} + \frac{(Mr)^\gamma}{R^\gamma}.$$

Choose $R = ((Mr)^{3(\gamma-1)} n/m)^{\frac{1}{3\gamma+1}}$.

D Proof of Theorem 1

Let t_m be the solution to $h_{t^*, m}(t) = 0$ as in Lemma 3.1. The consistency argument is immediate: the losses ℓ are convex, continuous, and locally Lipschitz, so Shapiro et al. [9, Thm. 5.4] gives $\hat{\theta}_n \xrightarrow{a.s.} \theta_L^*$. By an appeal to standard M-estimator theory [e.g. 11, Thm. 5.23], we thus obtain

$$\sqrt{n}(\hat{\theta}_n - \theta_L^*) \rightsquigarrow \mathcal{N} \left(0, \nabla^2 L(\theta_L^*, \sigma)^{-1} \text{Cov} \left(\frac{1}{m} \sum_{j=1}^m \nabla \ell_{\sigma, \theta_L^*}(Y_j | X) \right) \nabla^2 L(\theta_L^*, \sigma)^{-1} \right). \quad (7)$$

We expand the covariance term to obtain the first main result of the theorem. For shorthand, let $G_j = \nabla \ell_{\sigma, \theta_L^*}(Y_j | X)$, so that $\sum_{j=1}^m \mathbb{E}[G_j] = 0$ and the G_j are conditionally independent given X . Applying the law of total covariance, we have

$$\begin{aligned} \text{Cov} \left(\frac{1}{m} \sum_{j=1}^m G_j \right) &= \text{Cov} \left(\frac{1}{m} \sum_{j=1}^m \mathbb{E}[G_j | X] \right) + \mathbb{E} \left[\text{Cov} \left(\frac{1}{m} \sum_{j=1}^m G_j | X \right) \right] \\ &= \underbrace{\text{Cov} \left(\frac{1}{m} \sum_{j=1}^m \mathbb{E}[G_j | X] \right)}_{\text{(I)}} + \frac{1}{m^2} \sum_{j=1}^m \underbrace{\mathbb{E} [\text{Cov}(G_j | X)]}_{\text{(II)}}, \end{aligned}$$

where we have used the conditional independence of the Y_j conditional on X . We control each of terms (I) and (II) in turn.

For the first, we have by the independent decomposition $X = Zu^\star + P_{u^\star}^\perp W$ that

$$\mathbb{E}[G_j \mid X] = (\sigma(t_m Z)(1 - \sigma_j^\star(t^\star Z)) - \sigma(-t_m Z)\sigma_j^\star(t^\star Z)) X,$$

and so

$$\begin{aligned} \text{(I)} &= \mathbb{E} [(\sigma(t_m Z)(1 - \bar{\sigma}^\star(t^\star Z)) - \sigma(-t_m Z)\bar{\sigma}^\star(t^\star Z))^2 XX^\top] \\ &= \mathbb{E} [(\sigma(t_m Z)(1 - \bar{\sigma}^\star(t^\star Z)) - \sigma(-t_m Z)\bar{\sigma}^\star(t^\star Z))^2 Z^2] u^\star u^{\star\top} \\ &\quad + \mathbb{E} [(\sigma(t_m Z)(1 - \bar{\sigma}^\star(t^\star Z)) - \sigma(-t_m Z)\bar{\sigma}^\star(t^\star Z))^2] P_{u^\star}^\perp \Sigma P_{u^\star}^\perp \\ &= \mathbb{E}[\text{le}(Z)^2 Z^2] u^\star u^{\star\top} + \mathbb{E}[\text{le}(Z)^2] P_{u^\star}^\perp \Sigma P_{u^\star}^\perp, \end{aligned}$$

where we used the independence of W and Z . For term (II) above, we see that conditional on X , $\nabla \ell_{\sigma, \theta}(Y_j \mid X)$ is a binary random variable taking values in $\{-\sigma(-t_m Z)X, \sigma(t_m Z)X\}$ with probabilities $\sigma_j^\star(t^\star Z)$ and $1 - \sigma_j^\star(t^\star Z)$, respectively, so that a calculation leveraging the variance of Bernoulli random variables yields

$$\text{Cov}(G_j \mid X) = \sigma_j^\star(t^\star Z)(1 - \sigma_j^\star(t^\star Z)) (\sigma(t_m Z) + \sigma(-t_m Z))^2 XX^\top = v_j(Z)XX^\top,$$

where we used the definition (13) of the variance terms. A similar calculation to that we used for term (I) then gives that

$$\text{(II)} = \frac{1}{m^2} \sum_{j=1}^m \mathbb{E}[v_j(Z)Z^2] u^\star u^{\star\top} + \frac{1}{m^2} \sum_{j=1}^m \mathbb{E}[v_j(Z)] P_{u^\star}^\perp \Sigma P_{u^\star}^\perp.$$

Applying Lemma C.1 then allows us to decompose the the covariance in expression (7) into terms in the span of $u^\star u^{\star\top}$ and those perpendicular to it, so that the asymptotic covariance is

$$\frac{\mathbb{E}[\text{le}(Z)^2 Z^2] + \frac{1}{m^2} \sum_{j=1}^m \mathbb{E}[v_j(Z)Z^2]}{\mathbb{E}[\text{le}(Z)Z^2]^2} u^\star u^{\star\top} + \frac{\mathbb{E}[\text{le}(Z)^2] + \frac{1}{m^2} \sum_{j=1}^m \mathbb{E}[v_j(Z)]}{\mathbb{E}[\text{le}(Z)]^2} (P_{u^\star}^\perp \Sigma P_{u^\star}^\perp)^\dagger,$$

by applying Lemma 3.1 for the form of the Hessian $\nabla^2 L(\theta_L^\star, \sigma)$ and Lemma C.1 for the inverse, giving the first result of the theorem.

To obtain the second result, we apply the delta method with the mapping $\phi(x) = x / \|x\|_2$, which satisfies $\nabla\phi(x) = (I - \phi(x)\phi(x)^\top) / \|x\|_2$, so that for $\hat{u}_n = \hat{\theta}_n / \|\hat{\theta}_n\|_2$ we have

$$\sqrt{n}(\hat{u}_n - u^*) \rightsquigarrow \mathbf{N}\left(0, \frac{1}{t_m^2} \frac{\mathbb{E}[\text{le}(Z)^2] + \frac{1}{m^2} \sum_{j=1}^m \mathbb{E}[v_j(Z)]}{\mathbb{E}[\text{le}(Z)]^2} (\mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp)^\dagger\right)$$

as desired.

D.1 Proof of Lemma 3.1

Recall that $h = h_{t^*, m}$ is the calibration gap function (11). We see that because $\mathbb{E}[Z] = 0$, we have $h(0) = -2\mathbb{E}[Z\varphi_m(t^*Z)] < 0$, while using Assumption A3 and that $\mathbb{E}[|Z|] < \infty$, we apply dominated convergence and that $\lim_{t \rightarrow \infty} (\sigma(tZ) - \frac{1}{2})Z = c|Z|$ with probability 1 to obtain

$$\lim_{t \rightarrow \infty} h(t) = \mathbb{E}[c|Z|(1 - \varphi_m(t^*Z))] + \mathbb{E}[c|Z|\varphi_m(t^*Z)] = c\mathbb{E}[|Z|] > 0.$$

Because $h'(t) = \mathbb{E}[\sigma'(tZ)Z^2(1 - \varphi_m(t^*Z))] + \mathbb{E}[\sigma'(-tZ)Z^2\varphi_m(t^*Z)] > 0$, we see that there is a unique t_m solving $h(t_m) = 0$, and evidently $\theta_L^* = t_m u^*$ is a minimizer of L .

We compute the Hessian of L . For this, we again let $\theta = tu^*$ to write

$$\begin{aligned} \nabla^2 L(\theta, \sigma) &= \mathbb{E}[\sigma'(-\langle \theta, X \rangle) X X^\top \varphi_m(\langle X, \theta^* \rangle)] + \mathbb{E}[\sigma'(\langle \theta, X \rangle) X X^\top (1 - \varphi_m(\langle X, \theta^* \rangle))] \\ &= \mathbb{E}\left[\sigma'(-tZ)\varphi_m(t^*Z) \left(Z^2 u^* u^{*\top} + WW^\top\right)\right] + \mathbb{E}\left[\sigma'(tZ)(1 - \varphi_m(t^*Z)) \left(Z^2 u^* u^{*\top} + WW^\top\right)\right] \\ &= \mathbb{E}\left[(\sigma'(-tZ)\varphi_m(t^*Z) + \sigma'(tZ)(1 - \varphi_m(t^*Z))) Z^2\right] u^* u^{*\top} \\ &\quad + \mathbb{E}[\sigma'(-tZ)\varphi_m(t^*Z) + \sigma'(tZ)(1 - \varphi_m(t^*Z))] \mathbb{E}[WW^\top], \end{aligned}$$

which gives $\nabla^2 L(\theta, \sigma) \succ 0$ and so θ_L^* is unique. Finally, the desired form of the Hessian follows as $\mathbb{E}[WW^\top] = \mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp$.

E Proof of Theorem 2

As in the proof of Theorem 1, we begin with a consistency result. Let t_m be the solution to $h_{t^*,m}(t) = 0$ as in Lemma 3.1. The once again, Shapiro et al. [9, Thm. 5.4] shows that $\hat{\theta}_n \xrightarrow{a.s.} \theta_L^*$. As previously, appealing to standard M-estimator theory [e.g. 11, Thm. 5.23], we obtain

$$\sqrt{n}(\hat{\theta}_n - \theta_L^*) \rightsquigarrow \mathbf{N}(0, \nabla^2 L(\theta_L^*, \sigma)^{-1} \text{Cov}(\nabla \ell_{\sigma, \theta_L^*}(Y^+ | X)) \nabla^2 L(\theta_L^*, \sigma)^{-1}), \quad (8)$$

the difference from the asymptotic (7) appearing in the covariance term. Here, we recognize that conditional on $X = Zu^* + W$, the vector $\nabla \ell_{\sigma, \theta_L^*}(Y^+ | X)$ takes on the values $\{-\sigma(-t_m Z)X, \sigma(t_m Z)X\}$ each with probabilities $\varphi_m(t^* Z)$ and $1 - \varphi_m(t^* Z)$, respectively, while $\nabla \ell_{\sigma, \theta_L^*}(Y^+ | X)$ is (unconditionally) mean zero. Thus we have

$$\begin{aligned} \text{Cov}(\nabla \ell_{\sigma, \theta_L^*}(Y^+ | X)) &= \mathbb{E}[\sigma(-t_m Z)^2 \varphi_m(t^* Z) X X^\top + \sigma(t_m Z)^2 (1 - \varphi_m(t^* Z)) X X^\top] \\ &= \mathbb{E}[(\sigma(-t_m Z)^2 \varphi_m(t^* Z) + \sigma(t_m Z)^2 (1 - \varphi_m(t^* Z))) Z^2] u^* u^{*\top} \\ &\quad + \mathbb{E}[\sigma(-t_m Z)^2 \varphi_m(t^* Z) + \sigma(t_m Z)^2 (1 - \varphi_m(t^* Z))] P_{u^*}^\perp \Sigma P_{u^*}^\perp. \end{aligned}$$

Applying Lemma C.1 as in the proof of Theorem 1 to decompose the covariance terms in the asymptotic (8), and substituting in $\rho_m(t) = \sigma(t)\mathbb{1}\{t \geq 0\} + (1 - \sigma(t))\mathbb{1}\{t < 0\}$, the limiting covariance in expression (8) becomes

$$\begin{aligned} &\frac{\mathbb{E}[(\sigma(-t_m | Z|)^2 \rho_m(t^* Z) + \sigma(t_m | Z|)^2 (1 - \rho_m(t^* Z))) Z^2]}{\mathbb{E}[\text{he}(Z) Z^2]^2} u^* u^{*\top} \\ &+ \frac{\mathbb{E}[\sigma(-t_m | Z|)^2 \rho_m(t^* Z) + \sigma(t_m | Z|)^2 (1 - \rho_m(t^* Z))]}{\mathbb{E}[\text{he}(Z) Z^2]^2} (P_{u^*}^\perp \Sigma P_{u^*}^\perp)^\dagger. \end{aligned}$$

Lastly, we apply the delta method to $\phi(x) = x / \|x\|_2$, exactly as in the proof of Theorem 1, which gives the theorem.

F Proofs of asymptotic normality

In this appendix, we include proofs of the convergence results in Propositions 1, 2, and 3.

In each, we divide the proof into three steps: we characterize the loss minimizer, apply one of the master Theorems 1 or 2 to obtain asymptotic normality, and then characterize the behavior of the asymptotic covariance as $m \rightarrow \infty$.

F.1 Proof of Proposition 1

Asymptotic normality of the MLE. The asymptotic normality result is an immediate consequence of the classical asymptotics for maximum likelihood estimators [11, Thm. 5.29].

Normalized estimator. For the normalized estimator, we appeal to the master results developed in Section 3.1. In particular, since we are in the well-specified logistic model, we can invoke Corollary 3 and write directly that

$$\sqrt{n}(\hat{u}_{n,m}^{\text{lr}} - u^*) \rightsquigarrow \mathbf{N}\left(0, \frac{1}{m} \cdot \frac{1}{t^{\star 2}} \frac{\mathbb{E}[\sigma^{\text{lr}}(t^{\star}Z)(1 - \sigma^{\text{lr}}(t^{\star}Z))]}{\mathbb{E}[\sigma^{\text{lr}'}(t^{\star}Z)]^2} \mathbf{P}_{u^*}^{\perp} \Sigma \mathbf{P}_{u^*}^{\perp}\right),$$

which immediately implies

$$C(t) = \frac{1}{t^2} \frac{\mathbb{E}[\sigma^{\text{lr}}(tZ)(1 - \sigma^{\text{lr}}(tZ))]}{\mathbb{E}[\sigma^{\text{lr}'}(tZ)]^2} = \frac{1}{t^2 \mathbb{E}\left[\frac{e^{tZ}}{(1+e^{tZ})^2}\right]},$$

and that further $C(t)t^{2-\beta} = t^{-\beta} \mathbb{E}\left[\frac{e^{t|Z|}}{(1+e^{t|Z|})^2}\right]^{-1}$. To compute the limit when $t \rightarrow \infty$, we invoke Lemma C.2 and we conclude that

$$\lim_{t \rightarrow \infty} C(t)t^{2-\beta} = \lim_{t \rightarrow \infty} \frac{1}{t^{\beta} \mathbb{E}\left[\frac{e^{t|Z|}}{(1+e^{t|Z|})^2}\right]} = \frac{1}{c_Z \int_0^{\infty} \frac{z^{\beta-1} e^z}{(1+e^z)^2} dz}.$$

F.2 Proof of Proposition 2

Minimizer of the population loss. We can see identity (7) still holds with the calibration gap $h(t) = h_m(t) = \mathbb{E}[|Z|(1 - \rho_m(t^{\star}|Z|)) - \mathbb{E}\left[\frac{|Z|}{1+e^{t|Z|}}\right]]$ in Eq. (6) as $X - u^{\star}Z$ and $u^{\star}Z$ are indepen-

dent. The function $h(t)$ is monotonically increasing in t with $h(\infty) = \mathbb{E}[|Z|(1 - \rho_m(t^*|Z|))] > 0$, while $1 - \rho_m(t|Z|) \leq 1 - \rho_1(t|Z|) = \frac{1}{1+e^{t|Z|}}$, we must have $h(t^*) \leq 0$. Therefore there must be a unique zero point $t_m \geq t^*$ of $h(t)$, and so $t_m u^*$ is the unique minimizer of the population loss $L_m^{\text{mv}}(\theta)$.

Asymptotic variance. As t_m solves $h_m(t_m) = 0$, Eq. (7) guarantees that $t_m u^*$ is the global minimizer of the population loss L_m^{mv} . Appealing to Theorem 2, it follows that $\hat{\theta}_{n,m}^{\text{mv}} \xrightarrow{p} t_m u^*$, and

$$\sqrt{n}(\hat{u}_{n,m}^{\text{mv}} - u^*) \rightsquigarrow \mathbf{N}\left(0, C_m(t^*) (\mathbf{P}_{u^*} \Sigma \mathbf{P}_{u^*})^\dagger\right)$$

for the variance function (14), which in this case simplifies to

$$C_m(t^*) = \frac{\mathbb{E}\left[\frac{1}{(1+e^{t_m|Z|})^2} \rho_m(t^*|Z|) + \frac{1}{(1+e^{-t_m|Z|})^2} (1 - \rho_m(t^*|Z|))\right]}{t_m^2 \mathbb{E}\left[\frac{e^{t_m Z}}{(1+e^{t_m Z})^2}\right]^2}$$

via the symmetry $\rho_m(t) = \rho_m(-t)$.

Large m behavior. The remainder of the proof is to characterize the behavior of $C_m(t^*)$ as $m \rightarrow \infty$. We first derive asymptotics for t_m . To simplify notation, we let $\|\theta\|_2 = t = t^*$. Because t_m solves $h_m(t_m) = 0$ we have

$$\mathbb{E}\left[\frac{|Z|}{1 + e^{t_m|Z|}}\right] = \mathbb{E}[|Z|(1 - \rho_m(t|Z|))], \quad (9)$$

we must have $t_m \rightarrow \infty$ as $m \rightarrow \infty$ because $\rho_m(t) \rightarrow 1$ for any $t > 0$ as $m \rightarrow \infty$, so the right side of equality (9) converges to 0 by the dominated convergence theorem, and hence so must the left hand side. Invoking Lemma C.2 for the left hand side, it follows that

$$\lim_{m \rightarrow \infty} t_m^{\beta+1} \mathbb{E}\left[\frac{|Z|}{1 + e^{t_m|Z|}}\right] = \lim_{m \rightarrow \infty} t_m^\beta \mathbb{E}\left[\frac{t_m|Z|}{1 + e^{t_m|Z|}}\right] = c_Z \int_0^\infty \frac{z^\beta}{1 + e^z} dz,$$

while invoking Lemma C.3 for the right hand side of (9), it follows that

$$\begin{aligned}\lim_{m \rightarrow \infty} m^{\frac{\beta+1}{2}} \mathbb{E}[|Z|(1 - \rho_m(t|Z|))] &= \lim_{m \rightarrow \infty} m^{\frac{\beta}{2}} \mathbb{E}[\sqrt{m}|Z|(1 - \rho_m(t|Z|))] \\ &= c_Z \int_0^\infty z^\beta \Phi\left(-\frac{tz}{2}\right) dz = c_Z t^{-\beta-1} \int_0^\infty z^\beta \Phi\left(-\frac{z}{2}\right) dz,\end{aligned}$$

where the last line follows from change of variables $tz \mapsto z$. The identity (9) implies that the ratio $\mathbb{E}[|Z|/(1 + e^{t_m|Z|})]/\mathbb{E}[|Z|(1 - \rho_m(t|Z|))] = 1$ and so we have that as $m \rightarrow \infty$,

$$\frac{t_m}{\sqrt{m}} = \left(\frac{t_m^{\beta+1}}{m^{\frac{\beta+1}{2}}}\right)^{\frac{1}{\beta+1}} = \left(\frac{t_m^{\beta+1} \mathbb{E}\left[\frac{|Z|}{1+e^{t_m|Z|}}\right]}{m^{\frac{\beta+1}{2}} \mathbb{E}[|Z|(1 - \rho_m(t|Z|))]\right)^{\frac{1}{\beta+1}} \rightarrow \left(\frac{\int_0^\infty \frac{z^\beta}{1+e^z} dz}{\int_0^\infty z^\beta \Phi\left(-\frac{z}{2}\right) dz}\right)^{\frac{1}{\beta+1}} \cdot t =: at.$$

In particular, $t_m/t^* \sqrt{m} = a(1 + o_m(1))$.

We finally proceed to compute asymptotic behavior of $C_m(t^*)$, the variance (14). By Lemma C.2 the limit of its denominator as $t_m \rightarrow \infty$ satisfies

$$\lim_{m \rightarrow \infty} \underbrace{t_m^{2\beta} \mathbb{E}\left[\frac{e^{t_m Z}}{(1 + e^{t_m Z})^2}\right]^2}_{:=\text{den}(C_m(t))} = \lim_{t \rightarrow \infty} \left(t^\beta \mathbb{E}\left[\frac{e^{tZ}}{(1 + e^{tZ})^2}\right]\right)^2 = \left(c_Z \int_0^\infty \frac{z^{\beta-1} e^z}{(1 + e^z)^2} dz\right)^2.$$

We decompose the numerator into the two parts

$$\begin{aligned}& m^{\frac{\beta}{2}} \mathbb{E}\left[\frac{1}{(1 + e^{t_m|Z|})^2} \rho_m(t|Z|) + \frac{1}{(1 + e^{-t_m|Z|})^2} (1 - \rho_m(t|Z|))\right] \\ &= \underbrace{m^{\frac{\beta}{2}} \mathbb{E}\left[\left(\frac{1}{(1 + e^{-t_m|Z|})^2} - \frac{1}{(1 + e^{t_m|Z|})^2}\right) (1 - \rho_m(t|Z|))\right]}_{\text{(I)}} + \underbrace{m^{\frac{\beta}{2}} \mathbb{E}\left[\frac{1}{(1 + e^{t_m|Z|})^2}\right]}_{\text{(II)}}.\end{aligned}$$

As we have already shown that $m^{-\frac{1}{2}} t_m \rightarrow at$, we know for any $\epsilon > 0$ that for large enough m , $(1 - \epsilon)at\sqrt{m} \leq t_m \leq (1 + \epsilon)at\sqrt{m}$. We can thus invoke Lemma C.2 to get

$$\lim_{m \rightarrow \infty} \text{(II)} = \lim_{m \rightarrow \infty} m^{\frac{\beta}{2}} \mathbb{E}\left[\frac{1}{(1 + e^{\sqrt{m}at|Z|})^2}\right] = c_Z \int_0^\infty \frac{z^{\beta-1}}{(1 + e^{atz})^2} dz = c_Z t^{-\beta} \int_0^\infty \frac{z^{\beta-1}}{(1 + e^{az})^2} dz.$$

With the same argument, we apply Lemma C.3 to establish the convergence

$$\lim_{m \rightarrow \infty} \text{(I)} = c_Z \int_0^\infty z^{\beta-1} \left(\frac{1}{(1 + e^{-atz})^2} - \frac{1}{(1 + e^{atz})^2}\right) \Phi\left(-\frac{tz}{2}\right) dz$$

$$= c_Z \int_0^\infty z^{\beta-1} \frac{e^{atz} - 1}{e^{atz} + 1} \Phi\left(-\frac{tz}{2}\right) dz = c_Z t^{-\beta} \int_0^\infty z^{\beta-1} \frac{e^{az} - 1}{e^{az} + 1} \Phi\left(-\frac{z}{2}\right) dz,$$

where we use the change of variables $tz \mapsto z$. Taking limits, we have

$$\begin{aligned} \lim_{m \rightarrow \infty} m^{1-\frac{1}{2}\beta} C_m(t) &= \lim_{m \rightarrow \infty} \frac{m^{\frac{\beta}{2}} \mathbb{E}[\frac{1}{(1+e^{t_m|Z|})^2} \rho_m(t|Z|) + \frac{1}{(1+e^{-t_m|Z|})^2} (1 - \rho_m(t|Z|))] }{m^{\beta-1} t_m^{2-2\beta} \cdot t_m^{2\beta} \mathbb{E}[\frac{e^{t_m Z}}{(1+e^{t_m Z})^2}]^2} \\ &= \lim_{m \rightarrow \infty} \left(\frac{t_m}{\sqrt{m}} \right)^{2\beta-2} \cdot \frac{\lim_{m \rightarrow \infty} \text{(I)} + \lim_{m \rightarrow \infty} \text{(II)}}{\text{den}(C_m(t))} \\ &= (at)^{2\beta-2} \cdot \frac{c_Z t^{-\beta} \int_0^\infty z^{\beta-1} \left(\frac{1}{(1+e^{az})^2} + \frac{e^{az}-1}{e^{az}+1} \Phi\left(-\frac{z}{2}\right) \right) dz}{(c_Z \int_0^\infty \frac{z^{\beta-1} e^z}{(1+e^z)^2} dz)^2}, \end{aligned}$$

where we used that $t_m/\sqrt{m} \rightarrow at$ as above.

F.3 Proof of Proposition 3

Minimizer of the population loss. By Lemma 3.1, we know the gap $h(t) = 0$ has a unique solution t_m , with $t_m u^*$ minimizing the population loss $L(\theta, \sigma)$.

Asymptotic variance. Directly invoking Theorem 2 yields asymptotic normality:

$$\sqrt{n} (\hat{u}_{n,m}^{\text{mv}} - u^*) \rightsquigarrow \mathbf{N}\left(0, C_m(t^*) (\mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp)^\dagger\right),$$

where the covariance function (14) has the form

$$C_m(t^*) = \frac{1}{t_m^2} \frac{\mathbb{E}[\sigma(-t_m|Z|)^2 \rho_m(t^*Z) + \sigma(t_m|Z|)^2 (1 - \rho_m(t^*Z))]}{\mathbb{E}[\sigma'(t_m Z)]^2} \quad (10)$$

and again t_m is the implicitly defined zero of $h(t) = 0$.

Large m behavior. We derive the large m asymptotics of t_m and C_m under Assumption A4 and using the shorthand $\|\theta^*\|_2 = t$. The proof is essentially identical to that of Proposition 2 in Appendix F.2. First, recalling the probability (16), $\rho_m(t) = \mathbb{P}(Y^+ = \text{sign}(\langle X, \theta^* \rangle) \mid \langle X, \theta^* \rangle = t)$, we see that $1 - \rho_m(tz) \rightarrow 0$ for any $z \neq 0$ as $m \rightarrow \infty$, and thus by dominated convergence,

$\mathbb{E}[|Z|(1 - \rho_m(tZ))] \rightarrow 0$. The analogue of the identity (9) in the proof of Proposition 2, that t_m is the zero of $h(t) = \mathbb{E}[\sigma(t|Z)|Z|(1 - \rho_m(t^*Z))] - \mathbb{E}[\sigma(-t|Z)|Z|\rho_m(t^*Z)]$, implies

$$\mathbb{E}[\sigma(t_m|Z)|Z|(1 - \rho_m(t^*Z))] = \mathbb{E}[\sigma(-t_m|Z)|Z|\rho_m(t^*Z)]. \quad (11)$$

As σ is bounded and $\mathbb{E}[\sigma(-t|Z)|Z|\rho_m(t^*Z)] \rightarrow \mathbb{E}[\sigma(-t|Z)|Z|]$ for any t , the convergence of the left hand side of equality (11) to 0 as $m \rightarrow \infty$ means we must have $t_m \rightarrow \infty$. Invoking Lemma C.2 yields

$$\lim_{m \rightarrow \infty} t_m^{\beta+1} \mathbb{E}[|Z|\sigma(-t_m|Z)] = \lim_{m \rightarrow \infty} t_m^\beta \mathbb{E}[t_m|Z|\sigma(-t_m|Z)] = c_Z \int_0^\infty z^\beta \sigma(-z) dz.$$

Applying Lemma C.3 gives

$$m^{\frac{\beta+1}{2}} \mathbb{E}[|Z|(1 - \rho_m(tZ))] = m^{\frac{\beta}{2}} \mathbb{E}[\sqrt{m}|Z|(1 - \rho_m(tZ))] \rightarrow c_Z t^{-\beta-1} \int_0^\infty z^\beta \Phi(-2\bar{\sigma}^{*'}(0)z) dz.$$

Rewriting the identity (11) using the symmetry of σ , so that $\sigma(t) + \sigma(-t) = 1$, we have the equivalent statement that $\mathbb{E}[|Z|(1 - \rho_m(t^*Z))] = \mathbb{E}[\sigma(-t_m|Z)|Z|]$, or $\mathbb{E}[\sigma(-t_m|Z)|Z|]/\mathbb{E}[|Z|(1 - \rho_m(t^*Z))] = 1$. Using this identity ratio, we find that

$$\frac{t_m}{\sqrt{m}} = \left(\frac{t_m^{\beta+1} \mathbb{E}[|Z|\sigma(-t_m|Z)]}{m^{\frac{\beta+1}{2}} \mathbb{E}[|Z|(1 - \rho_m(tZ))]} \right)^{\frac{1}{\beta+1}} \rightarrow \left(\frac{\int_0^\infty z^\beta \sigma(-z) dz}{\int_0^\infty z^\beta \Phi(-2\bar{\sigma}^{*'}(0)z) dz} \right)^{\frac{1}{\beta+1}} \cdot t^* =: at^*.$$

This concludes the asymptotic characterization that $t_m = \sqrt{m}at^* \cdot (1 + o_m(1))$.

Finally, we turn to the asymptotics for $C_m(t^*)$ in (10). By Lemma C.2 its denominator has limit

$$\lim_{m \rightarrow \infty} \underbrace{t_m^{2\beta} \mathbb{E}[\sigma'(t_m Z)]^2}_{:= \text{den} C_m(t^*)} = \lim_{t \rightarrow \infty} (t^\beta \mathbb{E}[\sigma'(tZ)])^2 = \left(c_Z \int_0^\infty z^{\beta-1} \sigma'(z) dz \right)^2.$$

We decompose the (rescaled) numerator of the variance (10) into the two parts

$$\underbrace{m^{\frac{\beta}{2}} \mathbb{E}[(\sigma(t_m|Z))^2 - \sigma(-t_m|Z|^2)(1 - \rho_m(t|Z|))]}_{\text{(I)}} + \underbrace{m^{\frac{\beta}{2}} \mathbb{E}[\sigma(-t_m|Z|)^2]}_{\text{(II)}}.$$

Lemmas C.2 and that $t_m = a\sqrt{m}t^*(1 + o_m(1)) \rightarrow \infty$, coupled with the dominated convergence theorem, establishes the convergence

$$\lim_{m \rightarrow \infty} (\text{II}) = \lim_{m \rightarrow \infty} m^{\frac{\beta}{2}} \mathbb{E} [\sigma(-\sqrt{m}at^*|Z|)^2] = c_Z t^{-\beta} \int_0^\infty \frac{z^{\beta-1}}{(1 + e^{az})^2} dz.$$

Similarly, Lemma C.3 and that $t_m = a\sqrt{m}t^*(1 + o_m(1))$ gives that

$$\begin{aligned} \lim_{m \rightarrow \infty} (\text{I}) &= \lim_{m \rightarrow \infty} m^{\frac{\beta}{2}} \mathbb{E} [(\sigma(at^*\sqrt{m}|Z|)^2 - \sigma(-at^*\sqrt{m}|Z|)^2) (1 - \rho_m(tZ))] \\ &= c_Z \int_0^\infty z^{\beta-1} (\sigma(at^*z)^2 - \sigma(-at^*z)^2) \Phi(-2\bar{\sigma}'(0)tz) dz \\ &= c_Z t^{*\beta} \int_0^\infty z^{\beta-1} (\sigma(az)^2 - \sigma(-az)^2) \Phi(-2\bar{\sigma}'(0)z) dz, \end{aligned}$$

where in the last line we use change of variables $tz \mapsto z$. Hence we have

$$\begin{aligned} \lim_{m \rightarrow \infty} m^{1-\frac{1}{2}\beta} C_m(t^*) &= \lim_{m \rightarrow \infty} \frac{1}{m^{\beta-1} t_m^{2-2\beta}} \cdot \frac{\lim_{m \rightarrow \infty} (\text{I}) + \lim_{m \rightarrow \infty} (\text{II})}{\text{den}(C_m(t^*))} \\ &= \lim_{m \rightarrow \infty} \left(\frac{t_m}{\sqrt{m}} \right)^{2\beta-2} \cdot \frac{c_Z t^{*\beta} \int_0^\infty z^{\beta-1} (\sigma(az)^2 + (\sigma(az)^2 - \sigma(-az)^2) \Phi(-2\bar{\sigma}'(0)z)) dz}{(c_Z \int_0^\infty \frac{z^{\beta-1} e^z}{(1+e^z)^2} dz)^2}. \end{aligned}$$

Finally, as $t_m/\sqrt{m} = at^*(1 + o_m(1))$ we obtain that $m^{1-\beta/2} C_m(t^*) = t^{*\beta-2} b$ for some constant b depending only on $\beta, c_Z, \bar{\sigma}'(0)$, and σ .

G Proofs of fundamental limits of estimation with aggregate labels

G.1 Proof of Proposition 8

We assume without loss of generality m is odd. When m is even the proof is identical, except that we randomize Y^+ when we have equal votes for both classes. As the marginal distribution of X is $\mathcal{N}(0, I_d)$ for both $\mathbb{P}_{(X, Y^+)}^{\sigma^*, \theta^*, m}$ and $\mathbb{P}_{(X, Y^+)}^{\bar{\sigma}, \bar{\theta}, \bar{m}}$, we only have to show the existence

of a link $\bar{\sigma} \in \mathcal{F}_{\text{link}}^0$ such that the conditional distribution on any $X = x$ is the same, i.e.,

$$\mathbb{P}_{\sigma^*, \theta^*, m}(Y^+ = 1 \mid X = x) = \mathbb{P}_{\bar{\sigma}, \bar{\theta}, \bar{m}}(Y^+ = 1 \mid X = x).$$

For $m \in \mathbb{N}$, define the probability that a $\text{Binomial}(m, t)$ is at least $\lceil m/2 \rceil$ through the one-to-one transformation

$$P_m(t) := \sum_{i=\lceil m/2 \rceil}^m \binom{m}{i} t^i (1-t)^{m-i},$$

For any m , $P_m(t)$ monotonically increases in t , and $P_m(0) = 0, P_m(\frac{1}{2}) = \frac{1}{2}, P_m(1) = 1$, and by symmetry $P_m(t) + P_m(1-t) = 1$. By the definition of majority vote

$$\mathbb{P}_{\sigma^*, \theta^*, m}(Y^+ = 1 \mid X = x) = \sum_{i=\lceil m/2 \rceil}^m \binom{m}{i} \sigma^*(\langle \theta^*, x \rangle)^i (1 - \sigma^*(\langle \theta^*, x \rangle))^{m-i} = P_m \circ \sigma^*(\langle \theta^*, x \rangle).$$

Therefore, the link

$$\bar{\sigma}(t) := P_{\bar{m}}^{-1} \circ P_m \circ \sigma^* \left(\frac{\|\theta\|_2}{\|\bar{\theta}\|_2} t \right)$$

satisfies $\bar{\sigma}(0) = \frac{1}{2}$ and $\bar{\sigma}(t) - \frac{1}{2} > 0$ for all $t > 0$, since P_m maps $(\frac{1}{2}, 1]$ to $(\frac{1}{2}, 1]$. We also have

$$\begin{aligned} \bar{\sigma}(t) + \bar{\sigma}(-t) &= P_{\bar{m}}^{-1} \circ P_m \circ \sigma^* \left(\frac{\|\theta\|_2}{\|\bar{\theta}\|_2} t \right) + P_{\bar{m}}^{-1} \circ P_m \circ \sigma^* \left(-\frac{\|\theta\|_2}{\|\bar{\theta}\|_2} t \right) \\ &\stackrel{(i)}{=} P_{\bar{m}}^{-1} \circ P_m \circ \sigma^* \left(\frac{\|\theta\|_2}{\|\bar{\theta}\|_2} t \right) + P_{\bar{m}}^{-1} \circ P_m \circ \left(1 - \sigma^* \left(\frac{\|\theta\|_2}{\|\bar{\theta}\|_2} t \right) \right) \\ &\stackrel{(ii)}{=} P_{\bar{m}}^{-1} \circ P_m \circ \sigma^* \left(\frac{\|\theta\|_2}{\|\bar{\theta}\|_2} t \right) + P_{\bar{m}}^{-1} \circ \left(1 - P_m \circ \sigma^* \left(\frac{\|\theta\|_2}{\|\bar{\theta}\|_2} t \right) \right) \\ &= 1, \end{aligned}$$

where (i) and (ii) follow from symmetry of σ^* and P_m , respectively. Thus $\bar{\sigma}$ belongs to $\mathcal{F}_{\text{link}}^0$ and is a valid link function. This $\bar{\sigma}$ yields the desired equality:

$$\mathbb{P}_{\bar{\sigma}, \bar{\theta}, \bar{m}}(Y^+ = 1 \mid X = x) = P_{\bar{m}} \circ P_{\bar{m}}^{-1} \circ P_m \circ \sigma^* \left(\frac{\|\theta\|_2}{\|\bar{\theta}\|_2} \cdot \langle \bar{\theta}, x \rangle \right) = \mathbb{P}_{\sigma^*, \theta^*, m}(Y^+ = 1 \mid X = x).$$

G.2 Proof of Lemma 4.1

For $p_\theta(x, y) = \mathbb{P}_\theta(Y^+ = y \mid X = x)q(x)$, where q is the d -dimensional centered density of X , the Fisher information matrix at θ is

$$\mathbf{I}_m^{\text{mv}}(\theta) = \mathbb{E} \left[\nabla_\theta \log p_\theta(X, Y^+) \cdot \nabla_\theta \log p_\theta(X, Y^+)^\top \right].$$

Substituting $\rho_m(|\langle x, \theta^* \rangle|) = \mathbb{P}(Y^+ = \text{sign}(\langle x, \theta^* \rangle) \mid X = x)$, we then have

$$\begin{aligned} & \mathbb{E} \left[\nabla_\theta \log p_{\theta^*}(X, Y^+) \cdot \nabla_\theta \log p_{\theta^*}(X, Y^+)^\top \mid X = x \right] \\ &= \rho_m(|\langle x, \theta^* \rangle|) \cdot \frac{\nabla_\theta \rho_m(|\langle x, \theta^* \rangle|) \cdot \nabla_\theta \rho_m(|\langle x, \theta^* \rangle|)^\top}{\rho_m(|\langle x, \theta^* \rangle|)^2} \\ & \quad + (1 - \rho_m(|\langle x, \theta^* \rangle|)) \cdot \frac{(\nabla_\theta(1 - \rho_m(|\langle x, \theta^* \rangle|)) \cdot (\nabla_\theta(1 - \rho_m(|\langle x, \theta^* \rangle|)))^\top}{(1 - \rho_m(|\langle x, \theta^* \rangle|))^2} \\ &= \frac{\rho'_m(|\langle x, \theta^* \rangle|)^2}{\rho_m(|\langle x, \theta^* \rangle|)(1 - \rho_m(|\langle x, \theta^* \rangle|))} \cdot xx^\top \end{aligned}$$

as desired.

G.3 Proof of Theorem 3

Throughout this proof, we use $a_m \sim b_m$ to mean that $a_m/b_m \rightarrow 1$ as $m \rightarrow \infty$, or equivalently that $a_m = b_m \cdot (1 + o_m(1))$. By Lemma 4.1, we have

$$\mathbf{I}_m(\theta^*) = A_m(t^*) \cdot u^* u^{*\top} + B_m(t^*) \cdot \mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp$$

with

$$A_m(t) := \mathbb{E} \left[\frac{\rho'_m(t|Z|)^2 Z^2}{\rho_m(t|Z|)(1 - \rho_m(t|Z|))} \right], \quad (12a)$$

$$B_m(t) := \mathbb{E} \left[\frac{\rho'_m(t|Z|)^2}{\rho_m(t|Z|)(1 - \rho_m(t|Z|))} \right]. \quad (12b)$$

It is crucial to understand the behavior of ρ_m and ρ'_m , which have the exact forms

$$\rho_m(t) = \sum_{k > m/2} \binom{m}{k} \frac{e^{kt}}{(1 + e^t)^m},$$

$$\rho'_m(t) = \sum_{k>m/2} \binom{m}{k} \frac{ke^{kt}(1+e^t) - me^{(k+1)t}}{(1+e^t)^{m+1}}.$$

Throughout this proof, we assume without loss of generality that m is odd. When m is even, we replace $\binom{m}{\frac{m+1}{2}}$ with $\binom{m}{\frac{m}{2}+1}$, but otherwise all arguments are identical.

Lemma G.1. *The function ρ'_m is even, and for any $t > 0$,*

$$\rho'_m(t) = \binom{m}{\frac{m+1}{2}} \cdot \frac{m+1}{2} \frac{e^{\frac{m+1}{2}t}}{(1+e^t)^{m+1}}.$$

Proof. By direct calculations

$$\begin{aligned} \rho'_m(t) &= \sum_{k>m/2} \binom{m}{k} \frac{ke^{kt}(1+e^t) - me^{(k+1)t}}{(1+e^t)^{m+1}} \\ &\stackrel{(i)}{=} \sum_{k>m/2} \left\{ \binom{m}{k} \frac{ke^{kt}(1+e^t)}{(1+e^t)^{m+1}} - \binom{m}{k} \frac{ke^{(k+1)t}}{(1+e^t)^{m+1}} - \binom{m}{k+1} \frac{(k+1)e^{(k+1)t}}{(1+e^t)^{m+1}} \right\} \\ &= \frac{1}{(1+e^t)^{m+1}} \sum_{k>m/2} \left\{ \binom{m}{k} ke^{kt} - \binom{m}{k+1} (k+1)e^{(k+1)t} \right\}, \end{aligned}$$

where (i) uses the combinatorial identity

$$m \binom{m}{k} = k \binom{m}{k} + (k+1) \binom{m}{k+1}.$$

This yields a telescoping sum and establishes the result. $\square$

Next, we define the function

$$\eta_m(t) = \frac{\rho'_m(t)^2}{\rho_m(t)(1 - \rho_m(t))}.$$

We can then write for the Fisher information $\mathbf{l}_m(\theta^\star) = A_m(t^\star) \cdot u^\star u^{\star\top} + B_m(t^\star) \cdot \mathbf{P}_{u^\star}^\perp \Sigma \mathbf{P}_{u^\star}^\perp$ via

$$A_m(t) = \mathbb{E} [\eta_m(tZ)Z^2], \quad B_m(t) = \mathbb{E} [\eta_m(tZ)],$$

as η_m is an even function. The next lemma provides the key asymptotic guarantees for $\eta_m(t)$ in the above displays. See Appendix G.4 for a proof.

Lemma G.2. *There exists a function $c_m(t)$ satisfying the following:*

(i) *For any $t > 0$, $c_m(t) \geq 1 - e^{-t}$ and*

$$\eta_m(t) = \frac{(m+1)^2}{4} \binom{m}{\frac{m+1}{2}} \cdot \frac{e^{\frac{m+3}{2}t}}{(1+e^t)^{m+2}} \cdot c_m(t),$$

(ii) *For any $\delta \in (0, 1)$, there exist $C = C(\delta) > 0$ and $M = M(\delta)$ such that for all $m \geq M$*

and all $t > 0$,

$$c_m(t) \leq \frac{C \cdot (1 - e^{-t})}{1 - e^{-(\lfloor \delta \sqrt{m} \rfloor + 1)t}}.$$

(iii) *For any fixed $w > 0$,*

$$\lim_{m \rightarrow \infty} \sqrt{m} c_m \left(\frac{w}{\sqrt{m}} \right) = e^{-\frac{w^2}{8}} \sqrt{\frac{2}{\pi}} \cdot \frac{1}{\Phi(-w/2)\Phi(w/2)}.$$

We use the asymptotics in Lemma G.2 to compute $A_m(t)$ and $B_m(t)$, respectively.

Part I: Asymptotics for $B_m(t)$. We begin by showing the claimed asymptotic formula for $B_m(t)$ in the theorem's statement. By Stirling's formula, the multiplicative factor in η_m satisfies

$$\frac{(m+1)^2}{4} \binom{m}{\frac{m+1}{2}} \sim \frac{m^2}{4} \cdot 2^m \sqrt{\frac{2}{\pi m}},$$

and thus

$$\begin{aligned} B_m(t) &\sim \frac{m^2}{2\sqrt{2\pi m}} \mathbb{E} \left[2^m \frac{e^{\frac{m+3}{2}tZ}}{(1+e^{tZ})^{m+2}} \cdot c_m(tZ) \right] = \frac{m^2}{8\sqrt{2\pi m}} \mathbb{E} \left[2^{m+2} \frac{e^{\frac{m+3}{2}tZ}}{(1+e^{tZ})^{m+2}} \cdot c_m(tZ) \right] \\ &= \frac{m^2}{8\sqrt{2\pi m}} \mathbb{E} \left[\left(\frac{2e^{\frac{tZ}{2}}}{1+e^{tZ}} \right)^{m+2} \cdot e^{\frac{tZ}{2}} c_m(tZ) \right] \sim \underbrace{\frac{\sqrt{m}}{8\sqrt{2\pi}} \cdot (m+2) \mathbb{E} \left[\left(\frac{2e^{\frac{tZ}{2}}}{1+e^{tZ}} \right)^{m+2} \cdot e^{\frac{tZ}{2}} c_m(tZ) \right]}_{=: J_{m+2}(t)}. \end{aligned}$$

Our goal now is to show that

$$\lim_{m \rightarrow \infty} m^{\frac{\beta-1}{2}} J_m(t) = \frac{cZ}{t^\beta} \sqrt{\frac{8}{\pi}} \cdot \int_0^\infty \frac{e^{-z^2} z^{\beta-1}}{\Phi(-z)\Phi(z)} dz, \quad (13)$$

which then immediately implies the asymptotic $B_m(t) \sim bm(t\sqrt{m})^{-\beta}$ for $b = \frac{cZ}{4\pi} \int_0^\infty \frac{e^{-z^2} z^{\beta-1}}{\Phi(-z)\Phi(z)} dz$, as claimed in the theorem.

The proof of the limit (13) involves an argument via dominated convergence. By the change of variables $w := \sqrt{m}u$,

$$\begin{aligned} J_m(t) &= \int_0^\infty m \left(\frac{2e^{\frac{tu}{2}}}{1 + e^{tu}} \right)^m e^{\frac{tu}{2}} c_{m-2}(tu) p(u) du \\ &= \int_0^\infty \sqrt{m} \left(\frac{2e^{\frac{tw}{2\sqrt{m}}}}{1 + e^{\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} c_{m-2} \left(\frac{tw}{\sqrt{m}} \right) p \left(\frac{w}{\sqrt{m}} \right) dw \\ &= \left(\frac{1}{\sqrt{m}} \right)^{\beta-1} \int_0^\infty \left(\frac{2e^{\frac{tw}{2\sqrt{m}}}}{1 + e^{\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} w^{\beta-1} \cdot \sqrt{m} \left(\frac{w}{\sqrt{m}} \right)^{1-\beta} c_{m-2} \left(\frac{tw}{\sqrt{m}} \right) p \left(\frac{w}{\sqrt{m}} \right) dw. \end{aligned}$$

We now demonstrate dominating functions for the various terms in the above integrand. To bound the last several terms, we use Assumption A2 that $\kappa := \sup_{z \in (0, \infty)} z^{1-\beta} p(z) < \infty$ and Lemma G.2, which combine to give the upper bound

$$\begin{aligned} \sqrt{m} \left(\frac{w}{\sqrt{m}} \right)^{1-\beta} c_{m-2} \left(\frac{tw}{\sqrt{m}} \right) p \left(\frac{w}{\sqrt{m}} \right) &\leq \kappa \sqrt{m} c_{m-2} \left(\frac{tw}{\sqrt{m}} \right) \\ &\leq \frac{\kappa c \sqrt{m} \left(1 - e^{-\frac{tw}{\sqrt{m}}} \right)}{1 - e^{-(\lfloor \delta \sqrt{m-2} \rfloor + 1) \frac{tw}{\sqrt{m}}}} \leq c' \sup_{w \in (0, \infty)} \frac{tw}{1 - e^{-\delta tw}} < \infty \end{aligned}$$

for some $c' = c(\delta, \kappa)$ and all sufficiently large m . The next lemma controls the first terms in the integral above. (We defer its proof to Appendix G.5.)

Lemma G.3. *There exists a universal constant $0 < C$ such that for all m ,*

$$\left(\frac{2e^{\frac{tw}{2\sqrt{m}}}}{1 + e^{\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} w^\beta \leq \frac{1}{C} w^\beta \exp \left\{ -C \min \{ t^2 w^2, \sqrt{tw} \} \right\}.$$

Certainly the right hand side term in Lemma G.3 is integrable on $(0, \infty)$. Rewriting $J_m(t)$, we can therefore invoke dominated convergence to exchange the limit in m and integral to obtain

$$\lim_{m \rightarrow \infty} m^{\frac{\beta-1}{2}} J_m(t) = \int_0^\infty \lim_{m \rightarrow \infty} \left(\frac{2e^{\frac{tw}{2\sqrt{m}}}}{1 + e^{\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} w^{\beta-1} \cdot \sqrt{m} \left(\frac{w}{\sqrt{m}} \right)^{1-\beta} c_{m-2} \left(\frac{tw}{\sqrt{m}} \right) p \left(\frac{w}{\sqrt{m}} \right) dw$$

$$\begin{aligned}
&\stackrel{(i)}{=} c_Z \int_0^\infty \lim_{m \rightarrow \infty} w^{\beta-1} \exp \left\{ -\frac{t^2 w^2}{2} \cdot \left(\frac{\sqrt{m}}{tw} \cdot \left(1 - e^{-\frac{tw}{2\sqrt{m}}} \right) \right)^2 \right\} \cdot \lim_{m \rightarrow \infty} \sqrt{m} c_{m-2} \left(\frac{tw}{\sqrt{m}} \right) dw \\
&\stackrel{(ii)}{=} c_Z \int_0^\infty e^{-\frac{1}{8}t^2 w^2} \cdot e^{-\frac{t^2 w^2}{8}} \sqrt{\frac{2}{\pi}} \cdot \frac{w^{\beta-1}}{\Phi(-tw/2)\Phi(tw/2)} dw \\
&= \frac{c_Z}{t^\beta} \sqrt{\frac{8}{\pi}} \cdot \int_0^\infty \frac{e^{-z^2} z^{\beta-1}}{\Phi(-z)\Phi(z)} dz,
\end{aligned}$$

where in equality (i) we invoke Assumption A2 and in (ii) we use Lemma G.2. This gives the desired limit (13).

Part II: Asymptotics for $A_m(t)$. The calculations are similar to those we have done to approximate $B_m(t)$. In this case

$$A_m(t) \sim \frac{1}{8\sqrt{2\pi m}} \cdot (m+2)^2 \underbrace{\mathbb{E} \left[\left(\frac{2e^{\frac{tZ}{2}}}{1+e^{tZ}} \right)^{m+2} \cdot e^{\frac{tZ}{2}} c_m(tZ) Z^2 \right]}_{=: I_{m+2}(t)},$$

and again invoking dominated convergence,

$$\begin{aligned}
\lim_{m \rightarrow \infty} m^{\frac{\beta-1}{2}} I_m(t) &= \int_0^\infty \lim_{m \rightarrow \infty} \left(\frac{2e^{\frac{tw}{2\sqrt{m}}}}{1+e^{\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} w^{\beta-1} \cdot \sqrt{m} \left(\frac{w}{\sqrt{m}} \right)^{1-\beta} c_{m-2} \left(\frac{tw}{\sqrt{m}} \right) p \left(\frac{w}{\sqrt{m}} \right) w^2 dw \\
&= c_Z \int_0^\infty e^{-\frac{1}{8}t^2 w^2} \cdot e^{-\frac{t^2 w^2}{8}} \sqrt{\frac{2}{\pi}} \cdot \frac{w^{\beta+1}}{\Phi(-tw/2)\Phi(tw/2)} dw \\
&= \frac{8c_Z}{t^{\beta+2}} \sqrt{\frac{2}{\pi}} \cdot \int_0^\infty \frac{e^{-z^2} z^{\beta+1}}{\Phi(-z)\Phi(z)} dz.
\end{aligned}$$

Hence

$$A_m(t) = \frac{a}{t^2} \left(\frac{1}{t\sqrt{m}} \right)^\beta (1 + o_m(1)), \quad a = \frac{c_Z}{\pi} \int_0^\infty \frac{e^{-z^2} z^{\beta+1}}{\Phi(-z)\Phi(z)} dz.$$

G.4 Proof of Lemma G.2

We begin by proving part (i). Define

$$s_m(t) = \sum_{k < m/2} \binom{m}{k} e^{kt} \leq \frac{1}{2} (1 + e^t)^m,$$

where the inequality is valid for $t \geq 0$. Invoking Lemma G.1 yields

$$\begin{aligned}
\eta_m(t) &= \frac{\rho'_m(t)^2}{\rho_m(t)(1 - \rho_m(t))} = \frac{\frac{(m+1)^2}{4} \left(\frac{m}{2}\right)^2 \cdot \frac{e^{(m+1)t}}{(1+e^t)^{2m+2}}}{\left(\sum_{k>m/2} \binom{m}{k} e^{kt}\right) \left(\sum_{k<m/2} \binom{m}{k} e^{kt}\right) \cdot \frac{1}{(1+e^t)^{2m}}} \\
&= \frac{(m+1)^2}{4} \left(\frac{m}{2}\right)^2 \cdot \frac{e^{(m+1)t}}{(1+e^t)^{m+2}} \cdot \frac{1}{(1 - s_m(t)/(1+e^t)^m) s_m(t)} \\
&= \frac{(m+1)^2}{4} \left(\frac{m}{2}\right)^2 \cdot \frac{e^{\frac{m+3}{2}t}}{(1+e^t)^{m+2}} \cdot \underbrace{\frac{\left(\frac{m}{2}\right)^2 e^{\frac{m-1}{2}t}}{(1 - s_m(t)/(1+e^t)^m) s_m(t)}}_{:=c_m(t)},
\end{aligned}$$

which gives the equality in part (i). It then remains to show $c_m(t) \geq 1 - e^{-t}$. For this, we upper bound

$$e^{-\frac{m-1}{2}t} s_m(t) \leq \sum_{k<m/2} \binom{m}{k} e^{-(\frac{m-1}{2}-k)t} \leq \binom{m}{\frac{m+1}{2}} \frac{1}{1 - e^{-t}},$$

and noting the inequality $\frac{1}{2} \leq 1 - \frac{s_m(t)}{(1+e^t)^m} \leq 1$, it then follows that

$$c_m(t) \geq \frac{\left(\frac{m}{2}\right)^2 e^{\frac{m-1}{2}t}}{s_m(t)} \geq 1 - e^{-t}.$$

Proceeding to the proof of part (ii), for any $\delta \in (0, 1)$,

$$\begin{aligned}
\left(\frac{m}{\frac{m+1}{2}}\right)^{-1} e^{-\frac{m-1}{2}t} s_m(t) &= \sum_{l=0}^{\infty} \binom{m}{\frac{m+1}{2}-l}^{-1} e^{-lt} \geq \underbrace{\left(\frac{m}{\frac{m+1}{2}}\right)^{-1} \binom{m}{\frac{m+1}{2}-\lfloor \delta\sqrt{m} \rfloor}}_{:=q_m(\delta)} \sum_{l=0}^{\lfloor \delta\sqrt{m} \rfloor} e^{-lt} \\
&= q_m(\delta) \cdot \frac{1 - e^{-(\lfloor \delta\sqrt{m} \rfloor + 1)t}}{1 - e^{-t}},
\end{aligned}$$

where the inequality uses that $\binom{m}{\frac{m+1}{2}-l}$ is decreasing in l . We next establish that $q_m(\delta) \sim e^{-2\delta^2}$.

Indeed, applying Stirling's formula yields

$$\begin{aligned}
q_m(\delta) &= \frac{\left(\frac{m}{2}\right)^{\frac{m}{2}} \cdot \left(\frac{m}{2}\right)^{\frac{m}{2}}}{m^m} \cdot \frac{m^m}{\left(\frac{m}{2} - \delta\sqrt{m}\right)^{\frac{m}{2} - \delta\sqrt{m}} \cdot \left(\frac{m}{2} + \delta\sqrt{m}\right)^{\frac{m}{2} + \delta\sqrt{m}}} \cdot (1 + o_m(1)) \\
&= \left(1 - \frac{2\delta}{\sqrt{m}}\right)^{-\frac{m}{2} + \delta\sqrt{m}} \cdot \left(1 + \frac{2\delta}{\sqrt{m}}\right)^{-\frac{m}{2} - \delta\sqrt{m}} \cdot (1 + o_m(1)) \\
&= \left(1 - \frac{4\delta^2}{m}\right)^{-\frac{m}{2}} \cdot \left(1 - \frac{2\delta}{\sqrt{m}}\right)^{\delta\sqrt{m}} \cdot \left(1 + \frac{2\delta}{\sqrt{m}}\right)^{-\delta\sqrt{m}} \cdot (1 + o_m(1))
\end{aligned}$$

$$= e^{2\delta^2} \cdot e^{-2\delta^2} \cdot e^{-2\delta^2} \cdot (1 + o_m(1)) = e^{-2\delta^2} \cdot (1 + o_m(1)).$$

Substituting the above displays into the definition $c_m(t) = \binom{m}{\frac{m+1}{2}} e^{\frac{m-1}{2}t} s_m(t)^{-1} (1 - \frac{s_m(t)}{(1+e^t)^m})^{-1}$,

we can then establish the upper bound using $1 - s_m(t)/(1+e^t)^m \geq \frac{1}{2}$, so

$$c_m(t) \leq \frac{2 \binom{m}{\frac{m+1}{2}} e^{\frac{m-1}{2}t}}{s_m(t)} \leq \frac{2(1 - e^{-t})}{q_m(\delta) \cdot (1 - e^{-(\lfloor \delta\sqrt{m} \rfloor + 1)t})}.$$

Since $q_m(\delta) \sim e^{-2\delta^2}$, this inequality establishes part (ii).

For part (iii), we compute $\lim_{m \rightarrow \infty} \sqrt{m} c_m(w/\sqrt{m})$. For any fixed $\delta > 0$, we explicitly expand $s_m(w/\sqrt{m})$ to obtain

$$\begin{aligned} \left(\binom{m}{\frac{m+1}{2}} \right)^{-1} e^{-\frac{m-1}{2} \cdot \frac{w}{\sqrt{m}}} s_m(w/\sqrt{m}) &= \sum_{l=0}^{\infty} \left(\binom{m}{\frac{m+1}{2}} \right)^{-1} \binom{m}{\frac{m+1}{2} - l} e^{-lw/\sqrt{m}} \\ &= \underbrace{\sqrt{m} \sum_{k=0}^{\infty} \frac{1}{\sqrt{m}} \sum_{\lfloor k\delta\sqrt{m} \rfloor \leq l < \lfloor (k+1)\delta\sqrt{m} \rfloor} \left(\binom{m}{\frac{m+1}{2}} \right)^{-1} \binom{m}{\frac{m+1}{2} - l} e^{-lw/\sqrt{m}}}_{:= a_{m,k}(\delta)}. \end{aligned}$$

Here, we partition the set of all nonnegative integers $\{l \in \mathbb{N} \mid 0 \leq l < \infty\}$ into sets $L_k = \{\lfloor k\delta\sqrt{m} \rfloor \leq l < \lfloor (k+1)\delta\sqrt{m} \rfloor\}$ for $k = 0, 1, \dots$. Using the same asymptotics derived above for $q_m(\delta)$ and that $e^{-(k+1)\delta w} \leq e^{-lw/\sqrt{m}} \leq e^{-k\delta w}$ for all $l \in L_k$, it holds for some remainder term $r_{m,k}$ satisfying $|r_{m,k}(\delta)| \leq 1 - e^{-\delta w}$ that

$$\begin{aligned} a_{m,k}(\delta) &= \delta \cdot \frac{1}{\delta\sqrt{m}} \sum_{\lfloor k\delta\sqrt{m} \rfloor \leq l < \lfloor (k+1)\delta\sqrt{m} \rfloor} e^{-2k^2\delta^2} \cdot (1 + o_m(1)) \cdot e^{-lw/\sqrt{m}} \\ &= \delta \cdot \frac{1 + r_{m,k}(\delta)}{\delta\sqrt{m}} \sum_{\lfloor k\delta\sqrt{m} \rfloor \leq l < \lfloor (k+1)\delta\sqrt{m} \rfloor} e^{-2k^2\delta^2} \cdot (1 + o_m(1)) \cdot e^{-k\delta w} \\ &= (1 + r_{m,k}(\delta)) \cdot \delta e^{-2k^2\delta^2 - k\delta w} \cdot (1 + o_m(1)). \end{aligned}$$

We will invoke dominated convergence for series to allow us to exchange summation and limits in m . We observe the following termwise domination:

$$a_{m,k}(\delta) \leq \frac{1}{\sqrt{m}} \cdot (\lfloor (k+1)\delta\sqrt{m} \rfloor - \lfloor k\delta\sqrt{m} \rfloor) e^{-\lfloor k\delta\sqrt{m} \rfloor w/\sqrt{m}}$$

$$\leq \frac{\delta\sqrt{m} + 1}{\sqrt{m}} e^{-k\delta w + w/\sqrt{m}} \leq (\delta + 1)e^{-k\delta w + w},$$

which is summable in k . We can then invoke dominated convergence theorem for infinite

series to obtain that for some $|r(\delta)| \leq 1 - e^{-\delta w}$,

$$\lim_{m \rightarrow \infty} \sum_{k=0}^{\infty} a_{m,k}(\delta) = (1 + r(\delta)) \cdot \sum_{k=0}^{\infty} \delta e^{-2k^2\delta^2 - k\delta w},$$

implying that for any $\delta > 0$,

$$\left(\frac{m}{\frac{m+1}{2}}\right)^{-1} e^{-\frac{m-1}{2} \cdot \frac{w}{\sqrt{m}}} s_m(w/\sqrt{m}) \sim \sqrt{m} \cdot (1 + r(\delta)) \cdot \sum_{k=0}^{\infty} \delta e^{-2k^2\delta^2 - k\delta w}.$$

Since the left hand side does not depend on δ , we can then send δ to zero on the right hand

side and use the definition of Riemann integral to obtain

$$\begin{aligned} \left(\frac{m}{\frac{m+1}{2}}\right)^{-1} e^{-\frac{m-1}{2} \cdot \frac{w}{\sqrt{m}}} s_m(w/\sqrt{m}) &\sim \sqrt{m} \lim_{\delta \rightarrow 0} \sum_{k=0}^{\infty} \delta e^{-2k^2\delta^2 - k\delta w} \\ &= \sqrt{m} \int_0^{\infty} e^{-2x^2 - wx} dx = \sqrt{\frac{\pi m}{2}} e^{\frac{w^2}{8}} \Phi(-w/2). \end{aligned}$$

Now we compute the limit of the remaining term in $c_m(t)$,

$$\begin{aligned} s_m(w/\sqrt{m}) / (1 + e^{w/\sqrt{m}})^m &\sim \left(\frac{1}{1 + e^{w/\sqrt{m}}}\right)^m \cdot \left(\frac{m}{\frac{m+1}{2}}\right) e^{-w\sqrt{m}/2} \cdot \sqrt{\frac{\pi m}{2}} e^{\frac{w^2}{8}} \Phi(-w/2) \\ &\sim \left(\frac{2}{1 + e^{w/\sqrt{m}}}\right)^m \cdot e^{-w\sqrt{m}/2} e^{\frac{w^2}{8}} \Phi(-w/2) \\ &= \left(\frac{2e^{-\frac{w}{2\sqrt{m}}}}{1 + e^{\frac{w}{\sqrt{m}}}}\right)^m \cdot e^{\frac{w^2}{8}} \Phi(-w/2) = \left(1 - \frac{(1 - e^{-\frac{w}{2\sqrt{m}}})^2}{1 + e^{\frac{w}{\sqrt{m}}}}\right)^m \cdot e^{\frac{w^2}{8}} \Phi(-w/2) \\ &\sim \left(1 - \frac{1}{8} \frac{w^2}{m}\right)^m \cdot e^{\frac{w^2}{8}} \Phi(-w/2) \sim \Phi(-w/2). \end{aligned}$$

Taking the above displays collectively into

$$c_m(w/\sqrt{m}) = \frac{\left(\frac{m}{\frac{m+1}{2}}\right) e^{\frac{m-1}{2} \cdot \frac{w}{\sqrt{m}}}}{s_m(w/\sqrt{m})} \cdot \frac{1}{1 - s_m(w/\sqrt{m}) / (1 + e^{w/\sqrt{m}})^m}$$

we establish part (iii) of the lemma, that is,

$$\lim_{m \rightarrow \infty} \sqrt{m} c_m\left(\frac{w}{\sqrt{m}}\right) = \lim_{m \rightarrow \infty} \frac{\sqrt{m}}{\sqrt{\frac{\pi m}{2}} e^{\frac{w^2}{8}} \Phi(-w/2)} \cdot \frac{1}{1 - \Phi(-w/2)} = e^{-\frac{w^2}{8}} \sqrt{\frac{2}{\pi}} \cdot \frac{1}{\Phi(-w/2)\Phi(w/2)}.$$

G.5 Proof of Lemma G.3

We pick some constant $C > 0$. When $w \leq C\sqrt{m}/t$,

$$\begin{aligned}
& \left(\frac{2e^{\frac{tw}{2\sqrt{m}}}}{1 + e^{\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} w^\beta \\
&= \left(1 - \frac{\left(1 - e^{-\frac{tw}{2\sqrt{m}}}\right)^2}{1 + e^{-\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} w^\beta \leq e^C w^\beta \exp \left\{ -\frac{m \left(1 - e^{-\frac{tw}{2\sqrt{m}}}\right)^2}{1 + e^{-\frac{tw}{\sqrt{m}}}} \right\} \\
&\leq e^C w^\beta \exp \left\{ -\frac{1}{2} \cdot m \left(1 - e^{-\frac{tw}{2\sqrt{m}}}\right)^2 \right\} = e^C w^\beta \exp \left\{ -\frac{t^2 w^2}{2} \cdot \left(\frac{\sqrt{m}}{tw} \cdot \left(1 - e^{-\frac{tw}{2\sqrt{m}}}\right) \right)^2 \right\} \\
&\leq e^C w^\beta \cdot \exp \left\{ -\frac{t^2 w^2}{2} \cdot \left(\inf_{z \in (0, C]} \frac{1 - e^{-z/2}}{z} \right)^2 \right\},
\end{aligned}$$

and when $w > C\sqrt{m}/t$,

$$\begin{aligned}
\left(\frac{2e^{\frac{tw}{2\sqrt{m}}}}{1 + e^{\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} w^\beta &= \left(\frac{2e^{-\frac{tw}{2\sqrt{m}}}}{1 + e^{-\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} w^\beta \\
&\leq \left(2e^{-\frac{tw}{4\sqrt{m}}} \right)^m \cdot \exp \left\{ -\frac{\sqrt{tw}}{4\sqrt{m}} \cdot \sqrt{tw} \right\} e^{\frac{tw}{\sqrt{m}}} w^\beta \\
&\leq \left(2e^{-\frac{tw}{4\sqrt{m}} + \frac{tw}{m\sqrt{m}}} \right)^m \cdot w^\beta \exp \left\{ -\frac{C}{4} \sqrt{tw} \right\} \\
&\leq \left(2e^{-C(\frac{1}{4} - \frac{1}{m})} \right)^m w^\beta \exp \left\{ -\frac{C}{4} \sqrt{tw} \right\}.
\end{aligned}$$

Then for $m \geq 8$ and $C > 8 > \log 2$, it holds $2e^{-C(\frac{1}{4} - \frac{1}{m})} < 1$ and consequently

$$\left(\frac{2e^{\frac{tw}{2\sqrt{m}}}}{1 + e^{\frac{tw}{\sqrt{m}}}} \right)^m e^{\frac{tw}{\sqrt{m}}} w^\beta \leq w^\beta \exp \left\{ -\frac{C}{4} \sqrt{tw} \right\}.$$

G.6 Proof of Corollary 6

As we have the explicit formula for the density $p_\theta(X, Y^+) = \mathbb{P}(Y^+ | X)q(X)$, with $\mathbb{P}(Y^+ = \text{sign}(\langle x, \theta^* \rangle) | X = x) = \rho_m(|\langle x, \theta^* \rangle|)$ and q being the d -dimensional centered density with covariance Σ . It is differentiable in quadratic mean and allows us to invoke van der Vaart [11, Thm. 8.9]. It only remains to show the asymptotic rate in m for Σ_{θ^*} and Σ_{u^*} . Indeed,

by Lemma C.1,

$$\Sigma_{\theta^*} = \mathbf{I}_m(\theta^*)^{-1} = A_m(t^*)^{-1} \cdot u^* u^{*\top} + B_m(t^*)^{-1} \cdot (\mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp)^\dagger,$$

and Theorem 3 implies

$$\Sigma_{\theta^*} \asymp t^{*\beta+2} m^{\frac{\beta}{2}} \cdot u^* u^{*\top} + t^{*\beta} m^{\frac{\beta-2}{2}} \cdot (\mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp)^\dagger.$$

Analogously we establish the proof for Σ_{u^*} .

H Semiparametric approaches

We first provide master results that allow semi-parametric estimation of the link functions.

Consider Lipschitz symmetric link functions, and define for $L > 0$

$$\mathcal{F}_{\text{link}}^L := \{\sigma \mid \sigma \not\equiv 1/2 \text{ is non-decreasing, symmetric, and } L\text{-Lipschitz continuous}\} \subset \mathcal{F}_{\text{link}}^0.$$

We will demonstrate both consistency and asymptotic normality under appropriate convergence and regularity assumptions for the link functions. We assume there is a (semiparametric) collection $\mathcal{F}_{\text{link}}^{\text{sp}} \subset (\mathcal{F}_{\text{link}}^L)^m$ of link functions of interest, which may coincide with $(\mathcal{F}_{\text{link}}^L)^m$ but may be smaller, making estimation easier. Recall the distance $d_{\mathcal{F}_{\text{link}}}$ in Eq. (17). We make the following assumption.

Assumption A1. *The links $\vec{\sigma}^* \in \mathcal{F}_{\text{link}}^{\text{sp}}$ are normalized so that $\mathbb{P}(Y_j = y \mid X = x) = \sigma_j^*(y \langle x, u^* \rangle)$, and the sequence $\{\vec{\sigma}_n\} \subset \mathcal{F}_{\text{link}}^{\text{sp}}$ is consistent:*

$$\|\vec{\sigma}_n(-Y \langle X, u^* \rangle) - \vec{\sigma}^*(-Y \langle X, u^* \rangle)\|_{L^2(\mathbb{P})} \xrightarrow{P} 0.$$

Additionally, the mapping $(\theta, \vec{\sigma}) \mapsto \nabla_\theta^2 L(\theta, \vec{\sigma})$ is continuous for $d_{\mathcal{F}_{\text{link}}}$ restricted to $\mathbb{R}^d \times \mathcal{F}_{\text{link}}^{\text{sp}}$ at $(u^, \vec{\sigma}^*)$.*

The continuity of $\nabla_{\theta}^2 L(\theta, \vec{\sigma})$ allows us to develop local asymptotic normality. To see we may expect the assumption to hold, we give reasonably simple conditions sufficient for it in Lemma H.1. We can then give the formal version of the master result in Theorem 4 as follows.

Theorem H.1. *Let Assumption A1 hold and assume $|Z| > 0$ with probability one and has nonzero and continuous density $p(z)$ on $(0, \infty)$. Let Assumption A1 hold and assume that $\mathbb{E}[\|X\|_2^4] < \infty$. Then $\sqrt{n}(\hat{\theta}_{n,m}^{\text{sp}} - u^*)$ is asymptotically normal, and the normalized estimator $\hat{u}_{n,m}^{\text{sp}} = \hat{\theta}_{n,m}^{\text{sp}} / \|\hat{\theta}_{n,m}^{\text{sp}}\|_2$ satisfies*

$$\sqrt{n} (\hat{u}_{n,m}^{\text{sp}} - u^*) \rightsquigarrow \mathbf{N} \left(0, C_{m, \vec{\sigma}^*} (\mathbf{P}_{u^*}^{\perp} \Sigma \mathbf{P}_{u^*}^{\perp})^{\dagger} \right).$$

Next we give the promised sufficient conditions allowing the continuity of $\nabla_{\theta}^2 L(\theta, \vec{\sigma})$ at $(u^*, \vec{\sigma}^*)$, including that the collection of links $\mathcal{F}_{\text{link}}^{\text{sp}}$ is sufficiently smooth or the data distribution is continuous enough. (See Appendix J.1 for a proof.)

Lemma H.1. *Let $d_{\mathcal{F}_{\text{link}}}$ be the distance (17). Let Assumption A1 hold, where $|Z| > 0$ with probability one, has nonzero and continuously differentiable density $p(z)$ on $(0, \infty)$ satisfying $\lim_{z \rightarrow s} z^2 p(z) = 0$ for $s \in \{0, \infty\}$. The mapping $(\theta, \vec{\sigma}) \mapsto \nabla_{\theta}^2 L(\theta, \vec{\sigma})$ is continuous for $d_{\mathcal{F}_{\text{link}}}$ restricted to $\mathbb{R}^d \times \mathcal{F}_{\text{link}}^{\text{sp}}$ at $(u^*, \vec{\sigma}^*)$ if $\mathbb{E}[\|X\|_2^4] < \infty$ and either of the following conditions holds:*

(C.1) *For any $\vec{\sigma} = (\sigma_1, \dots, \sigma_m) \in \mathcal{F}_{\text{link}}^{\text{sp}}$, σ'_j are Lipschitz continuous.*

(C.2) *X has continuous density on $\mathbb{R}^d$.*

H.1 A single index model

Our first example application of Theorem H.1 is to a single index model. We present a multi-phase estimator that first estimates the direction $u^* = \theta^* / \|\theta^*\|_2$, then uses this estimate

to find a (consistent) estimate of the link σ^* , which we can then substitute directly into Theorem H.1. We defer all proofs of this section to Appendix K, which also includes a few auxiliary results that we use to prove the results proper.

We present the the abstract convergence result for link functions first, considering a scenario where we have an initial guess u_n^{init} of the direction u^* , independent of $(X_i, Y_{ij})_{i \leq n, j \leq m}$, for example constructed via a small held-out subset of the data. We set

$$\vec{\sigma}_n = \operatorname{argmin}_{\vec{\sigma} \in \mathcal{F}_{\text{link}}^{\text{sp}}} \sum_{i=1}^n \sum_{j=1}^m (\sigma_j(\langle u_n^{\text{init}}, X_i \rangle) - Y_{ij})^2,$$

where $\mathcal{F}_{\text{link}}^{\text{sp}} \subset (\mathcal{F}_{\text{link}}^{\text{L}})^m$ and so it consists of nondecreasing $\mathbf{L}$ -Lipschitz link functions with $\sigma(0) = \frac{1}{2}$. We assume that for all n , there exists a (potentially random) ϵ_n such that

$$\|u_n^{\text{init}} - u^*\|_2 \leq \epsilon_n.$$

Proposition 9. *Let X_i be vectors with $\mathbb{E}[\|X\|_2^k] < \infty$, where $k \geq 2$. Then with probability 1, there is a finite (random) $C < \infty$ such that for all large enough n ,*

$$\|\vec{\sigma}_n(Y \langle u^*, X \rangle) - \vec{\sigma}^*(Y \langle u^*, X \rangle)\|_{L^2(\mathbb{P})}^2 \leq C \left[n^{\frac{2}{3k} - \frac{2}{3}} + \epsilon_n^2 + \mathbf{L} n^{-\frac{k}{2(k+1)}} \right].$$

The proof is more or less a consequence of standard convergence results for nonparametric function estimation, though we include it for completeness in Appendix K.2 as it includes a few additional technicalities because of the initial estimate of u^* .

Summarizing, we see that a natural procedure is available: if we have models powerful enough to accurately estimate the conditional label probabilities $Y \mid X$, then Proposition 9 coupled with Theorem H.1 shows that we can achieve estimation with near-optimal asymptotic covariance. In particular, if u_n^{init} is consistent (so $\epsilon_n \xrightarrow{p} 0$), then $\widehat{\theta}_{n,m}^{\text{sp}}$ induces a normalized estimator $\widehat{u}_n^{\text{sp}} = \widehat{\theta}_{n,m}^{\text{sp}} / \|\widehat{\theta}_{n,m}^{\text{sp}}\|_2$ satisfying $\sqrt{n}(\widehat{u}_n^{\text{sp}} - u^*) \rightsquigarrow \mathbf{N}(0, C_{m,\vec{\sigma}}(\mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp)^\dagger)$.

H.2 Crowdsourcing model

Crowdsourcing typically targets estimating rater reliability, then using these reliability estimates to recover ground truth labels as accurately as possible, with versions of this approach central since at least Dawid and Skene’s Expectation-Maximization-based approaches [2, 14, 8]. We focus here on a simple model of rater reliability, highlighting how—at least in our stylized model of classifier learning—by combining a crowdsourcing reliability model and still using all labels in estimating a classifier, we can achieve asymptotically efficient estimates of θ^* , rather than the robust but slower estimates $\hat{\theta}_{n,m}^{\text{mv}}$ arising from “cleaned” labels.

We adopt Whitehill et al.’s roughly “low-rank” model for label generation [14]: for binary classification with m labelers and distinct link functions σ_i^* , model the difficulty of X_i by $\beta_i \in (-\infty, \infty)$, where $\text{sign}(\beta_i)$ denotes the true class X_i belongs to. A parameter α_j models the expertise of annotator j , and the probability labeler j correctly classifies X_i is

$$\mathbb{P}(Y_{ij} = 1) = \frac{1}{1 + \exp(-\alpha_j \beta_i)}.$$

(See also Raykar et al. [8].) The focus in these papers was to construct gold-standard labels and datasets (X_i, Y_i) ; here, we take the alternative perspective we have so far advocated to show how using all labels can yield strong performance.

We thus adopt a semiparametric approach: we model the labelers, assuming a black-box crowdsourcing model that can infer each labeler’s ability, then fit the classifier. We represent labeler j ’s expertise by a scalar $\alpha_j^* \in (0, \infty)$. Given data $X_i = X$ and the normalized $\theta^* = u^*$, we assume a modified logistic link

$$\mathbb{P}(Y_{ij} = 1 \mid X_i = x) = \frac{1}{1 + \exp(-\alpha_j^* \langle \theta^*, x \rangle)} = \sigma^{\text{lr}}(\alpha_j^* \langle u^*, x \rangle),$$

so $\alpha_j^* = \infty$ represents an omniscient labeler while $\alpha_j^* = 0$ means the labeler chooses random labels regardless of the data. Let $\alpha^* = (\alpha_1^*, \dots, \alpha_m^*) \in \mathbb{R}_+^m$. Then, in keeping with the plug-in

approach of the master Theorem H.1, we assume the blackbox crowdsourcing model generates an estimate $\alpha_n = (\alpha_{n,1}, \dots, \alpha_{n,m}) \in \mathbb{R}_+^m$ of α^* from the data $\{(X_i, (Y_{i1}, \dots, Y_{im}))\}_{i=1}^n$.

We consider the algorithm using the blackbox crowdsourcing model and empirical risk minimization with the margin-based loss $\ell_{\vec{\sigma}_n, \theta}$ as in Section 6, with $\vec{\sigma}_n(t) := (\sigma^{\text{lr}}(\alpha_{n,1}t), \dots, \sigma^{\text{lr}}(\alpha_{n,m}t))$, or equivalently using the rescaled logistic loss

$$\ell_{\vec{\sigma}_n, \theta}(y | x) = \frac{1}{m} \sum_{j=1}^m \ell_{\theta}^{\text{lr}}(y_j | \alpha_{n,j}x).$$

This allows us to apply our general semiparametric Theorem H.1 as long as the crowdsourcing model produces a consistent estimate $\alpha_n \xrightarrow{p} \alpha^*$ (see Appendix J.2 for a proof):

Proposition 10. *Let Assumption A1 hold, $|Z| > 0$ have nonzero and continuous density $p(z)$ on $(0, \infty)$, and $\mathbb{E}[\|X\|_2^4] < \infty$. If $\alpha_n \xrightarrow{p} \alpha^* \in \mathbb{R}^m$, then $\sqrt{n}(\hat{\theta}_{n,m}^{\text{sp}} - u^*)$ is asymptotically normal, and the normalized estimator $\hat{u}_{n,m}^{\text{sp}} = \hat{\theta}_{n,m}^{\text{sp}} / \|\hat{\theta}_{n,m}^{\text{sp}}\|_2$ satisfies*

$$\sqrt{n}(\hat{u}_{n,m}^{\text{sp}} - u^*) \rightsquigarrow \text{N}\left(0, \frac{1}{\sum_{j=1}^m \mathbb{E}[\sigma^{\text{lr}}(\alpha_j^* Z)(1 - \sigma^{\text{lr}}(\alpha_j^* Z))]} (\mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp)^\dagger\right).$$

By Proposition 10, the semiparametric estimator $\hat{u}_{n,m}^{\text{sp}}$ is efficient when the rater reliability estimates $\alpha_n \in \mathbb{R}^m$ are consistent. It is also immediate that if $\alpha_j^* \leq \alpha_{\max} < \infty$ are bounded, then the asymptotic covariance multiplier $C_{m, \alpha^*} = (\sum_{j=1}^m \mathbb{E}[\sigma^{\text{lr}}(\alpha_j^* Z)(1 - \sigma^{\text{lr}}(\alpha_j^* Z))])^{-1} = O(1/m)$, so we recover the $1/m$ scaling of the MLE, as opposed to the slower rates of the majority vote estimators in Section 3.2. At this point, the refrain is perhaps unsurprising: using all the label information can yield much stronger convergence guarantees.

I Proof of Theorem H.1

We divide the theorem into two main parts, consistent with the typical division of asymptotic normality results into a consistency result and a distributional result. Recall the notation

$L(\theta, \vec{\sigma}) = \mathbb{E}[\ell_{\vec{\sigma}, \theta}(Y | X)]$ and $\|\vec{\sigma} - \vec{g}\|_{L^2(\mathbb{P})}^2 = \mathbb{E}[\|\vec{\sigma}(Z) - \vec{g}(Z)\|_2^2]$, where Z has the distribution that Assumption A1 specifies.

I.1 Proof of consistency

We demonstrate the consistency $\widehat{\theta}_{n,m}^{\text{sp}} \xrightarrow{P} u^*$ in three parts, which we present as Lemmas I.1, I.2, and I.4. The first presents an analogue of Lemma 3.1 generalized to the case in which there are m distinct link functions, allowing us to characterize the link-dependent minimizers

$$\theta_{\vec{\sigma}}^* := \underset{\theta}{\operatorname{argmin}} L(\theta, \vec{\sigma})$$

via a one-dimensional scalar on the line $\{tu^* \mid t \in \mathbb{R}_+\}$. The second, Lemma I.2, then shows that $\theta_{\vec{\sigma}}^* \rightarrow u^*$ as $\vec{\sigma}$ approaches $\vec{\sigma}^*$, where the scaling that $\|\theta_{\vec{\sigma}}^*\|_2 \rightarrow 1$ follows by the normalization Assumption A1. Finally, the third lemma demonstrates the probabilistic convergence $\widehat{\theta}_{n,m}^{\text{sp}} - \theta_{\vec{\sigma}_n}^* \xrightarrow{P} 0$ whenever $\vec{\sigma}_n \xrightarrow{P} \vec{\sigma}^*$ in $L^2(P)$.

We begin with the promised analogue of Lemma 3.1.

Lemma I.1. *Define the calibration gap function*

$$h_{\vec{\sigma}}(t) := \frac{1}{m} \sum_{j=1}^m \mathbb{E}[(\sigma_j(tZ) - \sigma_j^*(Z)) Z]. \quad (14)$$

Then the loss $L(\theta, \vec{\sigma})$ has unique minimizer $\theta_{\vec{\sigma}}^ = t_{\vec{\sigma}} u^*$ for the unique $t_{\vec{\sigma}} \in (0, \infty)$ solving $h_{\vec{\sigma}}(t) = 0$. Additionally, taking $\text{he}_j(t, z) := \sigma_j'(-tz)\sigma_j^*(z) + \sigma_j'(tz)\sigma_j^*(-z)$, we have*

$$\nabla^2 L(tu^*, \vec{\sigma}) = \frac{1}{m} \sum_{j=1}^m \left(\mathbb{E}[\text{he}_j(t, Z) Z^2] u^* u^{*\top} + \mathbb{E}[\text{he}_j(t, Z)] \mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp \right) \succ 0.$$

Proof. We perform a derivation similar to that we used to derive the gap function (11), with a few modifications to allow collections of m link functions. Note that

$$L(\theta, \vec{\sigma}) = \frac{1}{m} \sum_{j=1}^m \mathbb{E}[\ell_{\sigma_j, \theta}(1 | X) \sigma_j^*(\langle X, u^* \rangle) + \ell_{\sigma_j, \theta}(-1 | X) \sigma_j^*(-\langle X, u^* \rangle)],$$

so that leveraging the usual ansatz that $\theta = tu^*$, we have

$$\begin{aligned}\nabla L(\theta, \vec{\sigma}) &= \mathbb{E} \left[\frac{1}{m} \sum_{j=1}^m (\sigma_j^*(-\langle X, u^* \rangle) \sigma_j(\langle \theta, X \rangle) - \sigma_j^*(\langle X, u^* \rangle) \sigma_j(-\langle \theta, X \rangle)) X \right] \\ &= \frac{1}{m} \sum_{j=1}^m \mathbb{E} [(\sigma_j^*(-Z) \sigma_j(tZ) - \sigma_j^*(Z) \sigma_j(-tZ)) Z] u^* \\ &= h_{\vec{\sigma}}(t) u^*,\end{aligned}$$

where $h_{\vec{\sigma}}(t) = \frac{1}{m} \sum_{j=1}^m \mathbb{E}[(\sigma_j^*(-Z) \sigma_j(tZ) - \sigma_j^*(Z) \sigma_j(-tZ)) Z]$ is the immediate generalization of the gap (11). We now simplify it to the form (14). Using the symmetries $\sigma^*(z) = 1 - \sigma^*(-z)$ and $\sigma(z) = 1 - \sigma(-z)$ for any $\sigma, \sigma^* \in \mathcal{F}_{\text{link}}$, we observe that

$$\sigma(tz)(1 - \sigma^*(z)) - \sigma(-tz)\sigma^*(z) = \sigma(tz) - (\sigma(tz) + \sigma(-tz))\sigma^*(z) = \sigma(tz) - \sigma^*(z),$$

and so $h_{\vec{\sigma}}(t) = \frac{1}{m} \sum_{j=1}^m \mathbb{E}[(\sigma_j(tZ) - \sigma_j^*(Z)) Z]$. Of course, $h_{\vec{\sigma}}(0) = -\frac{1}{m} \sum_{j=1}^m \mathbb{E}[\sigma_j^*(Z) Z] < 0$, as $\mathbb{E}[Z] = 0$, while $\lim_{t \rightarrow \infty} h_{\vec{\sigma}}(t) = \frac{1}{m} \sum_{j=1}^m \mathbb{E}[|Z| - \sigma_j^*(Z) Z] > 0$. Then as $h'_{\vec{\sigma}}(t) = \frac{1}{m} \sum_{j=1}^m \mathbb{E}[\sigma'_j(tZ) Z^2] > 0$, there exists a unique $t_{\vec{\sigma}} \in (0, \infty)$ satisfying $h_{\vec{\sigma}}(t_{\vec{\sigma}}) = 0$.

We turn to the Hessian derivation, where as in the proof of Lemma 3.1, we write for $\theta = tu^*$ that

$$\begin{aligned}\nabla^2 L(\theta, \vec{\sigma}) &= \frac{1}{m} \sum_{j=1}^m \mathbb{E} [(\sigma'_j(-tZ) \sigma_j^*(Z) + \sigma'_j(tZ) \sigma_j^*(-Z)) Z^2] u^* u^{*\top} \\ &\quad + \frac{1}{m} \sum_{j=1}^m \mathbb{E} [(\sigma'_j(-tZ) \sigma_j^*(Z) + \sigma'_j(tZ) \sigma_j^*(-Z))] P_{u^*}^\perp \Sigma P_{u^*}^\perp,\end{aligned}$$

and so $\nabla^2 L(\theta, \vec{\sigma}) \succ 0$ and $L(\theta, \vec{\sigma})$ has unique minimizer $\theta_{\vec{\sigma}}^* = t_{\vec{\sigma}} u^*$. $\square$

With Lemma I.1 serving as the analogue of Lemma 3.1, we can now show the continuity of the optimizing parameter $\theta_{\vec{\sigma}}^*$ in $\vec{\sigma}$:

Lemma I.2. *As $\|\vec{\sigma} - \vec{\sigma}^*\|_{L^2(\mathbb{P})} \rightarrow 0$, we have $\theta_{\vec{\sigma}}^* - u^* \rightarrow 0$.*

Proof. Via Lemma I.1, it is evidently sufficient to show that the solution $t_{\vec{\sigma}}$ to $h_{\vec{\sigma}}(t) = 0$ converges to 1. To show this, note the expansion

$$mh_{\vec{\sigma}}(t) = \sum_{j=1}^m (\mathbb{E}[(\sigma_j(tZ) - \sigma_j^*(tZ))Z] + \mathbb{E}[(\sigma_j^*(tZ) - \sigma_j^*(Z))Z]). \quad (15)$$

We use the following claim, which shows that the first term tends to 0 uniformly in t near 1:

Claim I.3. *For any $\sigma, \sigma^* \in \mathcal{F}_{\text{link}}$, we have $\sup_{t \in [\frac{1}{2}, 2]} \|\sigma(tZ) - \sigma^*(tZ)\|_{L^2(\mathbb{P})} \rightarrow 0$ whenever $\|\sigma^* - \sigma\|_{L^2(\mathbb{P})} \rightarrow 0$.*

Proof. We use that the density $p(z)$ of $|Z|$ is continuous and nonzero on $(0, \infty)$. Take any $0 < M_0 < M_1 < \infty$. Then

$$\begin{aligned} \sup_{t \in [\frac{1}{2}, 2]} \|\sigma^*(tZ) - \sigma(tZ)\|_{L^2(\mathbb{P})} &= \sup_{t \in [\frac{1}{2}, 2]} \sqrt{\int_0^\infty (\sigma^*(tz) - \sigma(tz))^2 p(z) dz} \\ &\stackrel{(i)}{\leq} \mathbb{P}(|Z| \leq M_0) + \mathbb{P}(|Z| \geq M_1) + \sup_{t \in [\frac{1}{2}, 2]} \sqrt{\int_{M_0}^{M_1} (\sigma^*(tz) - \sigma(tz))^2 p(tz) \cdot \frac{p(z)}{p(tz)} dz} \\ &\leq \mathbb{P}(|Z| \leq M_0) + \mathbb{P}(|Z| \geq M_1) + \sup_{\frac{M_0}{2} \leq z, z' \leq 2M_1} \sqrt{\frac{p(z)}{p(z')}} \sup_{t \in [\frac{1}{2}, 2]} \sqrt{\int_{M_0}^{M_1} (\sigma^*(tz) - \sigma(tz))^2 p(tz) dz}, \end{aligned}$$

where in (i) we use that the link functions σ^* and σ are bounded within $[0, 1]$. For any fixed $0 < M_0 \leq M_1 < \infty$, the ratio $\frac{p(z)}{p(z')}$ is bounded for $z, z' \in [\frac{1}{2}M_0, 2M_1]$, and using the substitution $tz \mapsto z$ we have

$$\sup_{t \in [\frac{1}{2}, 2]} \sqrt{\int_{M_0}^{M_1} (\sigma^*(tz) - \sigma(tz))^2 p(tz) dz} \leq \sqrt{2} \cdot \|\sigma^* - \sigma\|_{L^2(\mathbb{P})}.$$

We thus have that $\|\sigma(tZ) - \sigma^*(tZ)\|_{L^2(\mathbb{P})} \leq K \|\sigma - \sigma^*\|_{L^2(\mathbb{P})} + \mathbb{P}(|Z| \notin [M_0, M_1])$, where K depends only on M_0, M_1 and the distribution of Z . Take $M_0 \downarrow 0$ and $M_1 \uparrow \infty$. $\square$

Leveraging the expansion (15) preceding Claim I.3 and the claim itself, we see that

$$h_{\vec{\sigma}}(t) = \mathbb{E}[Z^2] \cdot o(1) + \frac{1}{m} \sum_{j=1}^m \mathbb{E}[(\sigma_j^*(tZ) - \sigma_j^*(Z))Z]$$

uniformly in $t \in [\frac{1}{2}, 2]$ as $\|\vec{\sigma}^* - \vec{\sigma}\|_{L^2(\mathbb{P})} \rightarrow 0$. The monotonicity of each σ_j^* guarantees that if $f_j(t) = \mathbb{E}[(\sigma_j^*(tZ) - \sigma_j^*(Z))Z]$, then $f_j'(t) = \mathbb{E}[\sigma_j^{*\prime}(tZ)Z^2] > 0$, and so $t = 1$ uniquely solves $f_j(t) = 0$ and we must have $t_{\vec{\sigma}} \rightarrow 1$ as $\|\vec{\sigma} - \vec{\sigma}^*\|_{L^2(\mathbb{P})} \rightarrow 0$. $\square$

Finally, we proceed to the third part of the consistency argument: the convergence in probability.

Lemma I.4. *If $\|\vec{\sigma}_n - \vec{\sigma}\|_{L^2(\mathbb{P})} \xrightarrow{p} 0$, then $\widehat{\theta}_{n,m}^{\text{sp}} - \theta_{\vec{\sigma}_n}^* \xrightarrow{p} 0$.*

Proof. By Lemma I.1 and the assumed continuity of the population Hessian, there exists $\delta > 0$ such that

$$L(\theta, \vec{\sigma}) \geq L(\theta_{\vec{\sigma}}^*, \vec{\sigma}) + \frac{\lambda}{2} \|\theta - \theta_{\vec{\sigma}}^*\|_2^2 \quad (16)$$

whenever both $\|\vec{\sigma} - \vec{\sigma}^*\|_{L^2(\mathbb{P})} \leq \delta$ and $\|\theta_{\vec{\sigma}}^* - \theta\|_2 \leq \delta$. Applying the uniform convergence Lemma C.4, we see that for any $r < \infty$, we have

$$\sup_{\|\theta\|_2 \leq r, \vec{\sigma} \in \mathcal{F}_{\text{link}}^m} |P_n \ell_{\vec{\sigma}, \theta} - L(\theta, \vec{\sigma})| \xrightarrow{p} 0.$$

For $\delta > 0$, define the events

$$\mathcal{E}_n(\delta) := \left\{ \|\theta_{\vec{\sigma}}^* - u^*\|_2 \leq \delta, \|\vec{\sigma}_n - \vec{\sigma}^*\|_{L^2(\mathbb{P})} \leq \delta \right\},$$

where Lemma I.2 and the assumption that $\|\vec{\sigma}_n - \vec{\sigma}^*\|_{L^2(\mathbb{P})} \xrightarrow{p} 0$ imply that $\mathbb{P}(\mathcal{E}_n(\delta)) \rightarrow 1$ for all $\delta > 0$. By the growth condition (16) and uniform convergence $P_n \ell_{\vec{\sigma}, \theta} - L(\theta, \vec{\sigma}) \xrightarrow{p} 0$ over $\|\theta\|_2 \leq r$, we therefore have that with probability tending to 1,

$$\inf_{\|\theta - \theta_{\vec{\sigma}_n}^*\|_2 = \delta} \left\{ P_n \ell_{\theta, \vec{\sigma}_n} - P_n \ell_{\theta_{\vec{\sigma}_n}^*, \vec{\sigma}_n} \right\} \geq \frac{\lambda}{4} \delta^2.$$

The convexity of the losses $\ell_{\theta, \vec{\sigma}}$ in θ and that $\widehat{\theta}_{n,m}^{\text{sp}}$ minimizes $P_n \ell_{\theta, \vec{\sigma}_n}$ then guarantee the desired convergence $\widehat{\theta}_{n,m}^{\text{sp}} - \theta_{\vec{\sigma}_n}^* \xrightarrow{p} 0$. $\square$

I.2 Asymptotic normality via Donsker classes

While we do not have $\|\vec{\sigma}_n - \vec{\sigma}^*\|_{L^2(\mathbb{P})} = o_P(n^{-1/2})$, which allows the cleanest and simplest asymptotic normality results with nuisance parameters (e.g. [11, Thm. 25.54]), we still expect $\sqrt{n}(\hat{\theta}_{n,m}^{\text{sp}} - \theta_{\vec{\sigma}_n}^*)$ to be asymptotically normal, and therefore, as $\theta_{\vec{\sigma}}^* = tu^*$ for some $t > 0$, the normalized estimators $\hat{\theta}_{n,m}^{\text{sp}}/\|\hat{\theta}_{n,m}^{\text{sp}}\|_2$ should be asymptotically normal to u^* . To develop the asymptotic normality results, we perform an analysis of the empirical process centered at the *estimators* $\hat{\theta}_{n,m}^{\text{sp}}$, rather than the “true” parameter u^* as would be typical.

We begin with an expansion. We let $\theta_n^* = \theta_{\vec{\sigma}_n}^*$ for shorthand. Then as $\mathbb{P}_{n,m} \nabla_{\theta} \ell_{\hat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n} = 0$ and $\mathbb{P} \nabla_{\theta} \ell_{\theta_n^*, \vec{\sigma}_n} = 0$, we can derive

$$\begin{aligned} \mathbb{G}_{n,m} \nabla_{\theta} \ell_{\hat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n} &= \sqrt{n} \left(\mathbb{P}_{n,m} \nabla_{\theta} \ell_{\hat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n} - \mathbb{P} \nabla_{\theta} \ell_{\hat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n} \right) \\ &= \sqrt{n} \left(\mathbb{P} \nabla_{\theta} \ell_{\theta_n^*, \vec{\sigma}_n} - \mathbb{P} \nabla_{\theta} \ell_{\hat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n} \right) \\ &= \sqrt{n} \left(\nabla_{\theta} L(\theta_n^*, \vec{\sigma}_n) - \nabla_{\theta} L(\hat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n) \right) \\ &= \left(\int_0^1 \nabla_{\theta}^2 L((1-t)\hat{\theta}_{n,m}^{\text{sp}} + t\theta_n^*, \vec{\sigma}_n) dt \right) \cdot \sqrt{n}(\theta_n^* - \hat{\theta}_{n,m}^{\text{sp}}). \end{aligned}$$

The assumed continuity of $\nabla_{\theta}^2 L(\theta, \vec{\sigma})$ at $(u^*, \vec{\sigma}^*)$ (recall Assumption A1) then implies that

$$\begin{aligned} \sqrt{n}(\theta_n^* - \hat{\theta}_{n,m}^{\text{sp}}) &= \left(\int_0^1 \nabla_{\theta}^2 L((1-t)\theta_n^* + t\hat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n) dt \right)^{-1} \cdot \mathbb{G}_{n,m} \nabla_{\theta} \ell_{\hat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n} \\ &= \left(\nabla_{\theta}^2 L(u^*, \vec{\sigma}^*) + o_P(1) \right)^{-1} \cdot \mathbb{G}_{n,m} \nabla_{\theta} \ell_{\hat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n}, \end{aligned} \tag{17}$$

where we have used the consistency guarantees $\theta_n^* \xrightarrow{P} u^*$, $\hat{\theta}_{n,m}^{\text{sp}} \xrightarrow{P} u^*$ by Lemmas I.2 and I.4 and that $\|\vec{\sigma}_n - \vec{\sigma}^*\|_{L^2(\mathbb{P})} \xrightarrow{P} 0$ by assumption.

The expansion (17) forms the basis of our asymptotic normality result; while $\hat{\theta}_{n,m}^{\text{sp}}$ and $\vec{\sigma}_n$ may be data dependent, by leveraging uniform central limit theorems and the theory of Donsker function classes, we can show that $\mathbb{G}_{n,m} \nabla \ell$ has an appropriate normal limit. To that

end, define the function classes

$$\mathcal{F}_\delta := \{\nabla_\theta \ell_{\theta, \sigma} \mid \|\theta - u^\star\|_2 \leq \delta, \sigma \in \mathcal{F}_{\text{link}}\},$$

where we leave the Lipschitz constant $\mathbf{L}$ in $\mathcal{F}_{\text{link}}$ tacit. By Assumption A1 and that $\widehat{\theta}_{n,m}^{\text{sp}} \xrightarrow{p} u^\star$, we know with probability tending to 1 we have the membership $\nabla_\theta \ell_{\widehat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n} \in \mathcal{F}_\delta$. The key result is then that $\mathcal{F}_\delta$ is a Donsker class:

Lemma I.5. *Assume that $\mathbb{E}[\|X\|_2^4] < \infty$. Then $\mathcal{F}_\delta$ is a Donsker class, and moreover, if $d_{\mathcal{F}_{\text{link}}}((\widehat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n), (u^\star, \vec{\sigma}^\star)) \xrightarrow{p} 0$, then*

$$\mathbb{G}_{n,m} \nabla_\theta \ell_{\widehat{\theta}_{n,m}^{\text{sp}}, \vec{\sigma}_n} - \mathbb{G}_{n,m} \nabla_\theta \ell_{u^\star, \vec{\sigma}^\star} \xrightarrow{p} 0.$$

Temporarily deferring the proof of Lemma I.5, let us see how it leads to the proof of Theorem H.1. Using Lemma I.5 and Slutsky's lemmas in the equality (17), we obtain

$$\begin{aligned} \sqrt{n}(\widehat{\theta}_{n,m}^{\text{sp}} - \theta_n^\star) &= -(\nabla_\theta^2 L(u^\star, \vec{\sigma}^\star) + o_P(1))^{-1} \cdot \mathbb{G}_{n,m} \nabla_\theta \ell_{u^\star, \vec{\sigma}^\star} + o_P(1) \\ &\rightsquigarrow \mathbf{N}\left(0, \nabla^2 L(u^\star, \vec{\sigma}^\star) \text{Cov}\left(\frac{1}{m} \sum_{j=1}^m \nabla \ell_{u^\star, \sigma_j^\star}(Y_j \mid X)\right) \nabla^2 L(u^\star, \vec{\sigma}^\star)\right). \end{aligned}$$

Calculations completely similar to those we use in the proof of Theorem 1 then give Theorem H.1: because $\mathbb{P}(Y_j = y \mid X = x) = \sigma_j^\star(y \langle u^\star, x \rangle)$ by assumption,

$$\text{Cov}\left(\frac{1}{m} \sum_{j=1}^m \nabla \ell_{u^\star, \sigma_j^\star}(Y_j \mid X)\right) = \frac{1}{m^2} \sum_{j=1}^m \text{Cov}(\nabla \ell_{u^\star, \sigma_j^\star}(Y_j \mid X))$$

because $Y_j \mid X$ are conditionally independent, while

$$\begin{aligned} \text{Cov}(\nabla \ell_{u^\star, \sigma_j^\star}(Y_j \mid X)) &= \mathbb{E}[\sigma_j^\star(Z)(1 - \sigma_j^\star(Z))XX^\top] \\ &= \mathbb{E}[\sigma_j^\star(Z)(1 - \sigma_j^\star(Z))Z^2]u^\star u^{\star\top} + \mathbb{E}[\sigma_j^\star(Z)(1 - \sigma_j^\star(Z))]P_{u^\star}^\perp \Sigma P_{u^\star}^\perp. \end{aligned}$$

When $\sigma_j = \sigma_j^\star$, the Hessian function $\mathbf{h}_{e_j}$ in Lemma I.1 simplifies to $\mathbf{h}_{e_j}(1, z) = \sigma_j^{\star'}(z)$ as $\sigma_j^\star$ is symmetric about 0. We then apply the delta method as in the proof of Theorem 1.

Finally, we return to the proof of Lemma I.5.

Proof of Lemma I.5. To prove $\mathcal{F}_\delta$ is Donsker, we show that each coordinate of $\nabla_\theta \ell_{\theta,\sigma} \in \mathcal{F}_\delta$ is, and as $\nabla_\theta \ell_{\theta,\sigma}(y \mid x) = -y\sigma(-y\langle x, \theta \rangle)x$, this amounts to showing the coordinate functions $f_{\theta,\sigma}^{(i)}(v) = v_i\sigma(-\langle v, \theta \rangle)$ form a Donsker class when v has distribution $V = YX$. Let

$$\mathcal{F}_\delta^{(i)} := \left\{ f_{\theta,\sigma}^{(i)}(\cdot) \mid \|\theta - u^*\|_2 \leq \delta, \sigma \in \mathcal{F}_{\text{link}} \right\},$$

so it is evidently sufficient to prove that $\mathcal{F}_\delta^{(1)}$ forms a Donsker class.

We use bracketing and entropy numbers [12] to control the $\mathcal{F}_\delta^{(i)}$. Recall that for a function class $\mathcal{F}$, an ϵ -*bracket* of $\mathcal{F}$ in $L^q(\mathbb{P})$ is a collection of functions $\{(l_i, u_i)\}$ such that for each $f \in \mathcal{F}$, there exists i such that $l_i \leq f \leq u_i$ and $\|u_i - l_i\|_{L^q(\mathbb{P})} \leq \epsilon$. The *bracketing number* $N_{[]}(\epsilon, \mathcal{F}, L^q(\mathbb{P}))$ is the cardinality of the smallest such ϵ -bracket, and the *bracketing entropy* is

$$J_{[]}(\mathcal{F}, L^q(\mathbb{P})) := \int_0^\infty \sqrt{\log N_{[]}(\epsilon, \mathcal{F}, L^q(\mathbb{P}))} d\epsilon.$$

To show that $\mathcal{F}_\delta^{(i)}$ is Donsker, it is sufficient [12, Ch. 2.5.2] to show that $J_{[]}(\mathcal{F}_\delta^{(i)}, L^2(\mathbb{P})) < \infty$. Our approach to demonstrate that $\mathcal{F}_\delta^{(1)}$ is Donsker is thus to construct an appropriate ϵ -bracket of $\mathcal{F}_\delta^{(1)}$, which we do by first covering ℓ_2 -balls in $\mathbb{R}^d$, then for vectors $\theta \in \mathbb{R}^d$, constructing a bracketing of the induced function class $\{f_{\theta,\sigma}^{(1)}\}_{\sigma \in \mathcal{F}_{\text{link}}}$, which we combine to give the final bracketing of $\mathcal{F}_\delta^{(1)}$.

We proceed with this two-stage covering and bracketing. Let $\epsilon, \gamma > 0$ be small numbers whose values we determine later. Define the ℓ_2 ball $\mathbb{B}_2^d = \{x \in \mathbb{R}^d \mid \|x\|_2 \leq 1\}$, and for any $0 < \epsilon < \delta$, let $\mathcal{N}_\epsilon$ be a minimal ϵ -cover of $\delta\mathbb{B}_2^d$ in the Euclidean norm of size $N = N(\epsilon, \delta\mathbb{B}_2^d, \|\cdot\|_2)$, $\mathcal{N}_\epsilon = \{\theta_1, \dots, \theta_N\}$, so that for any θ with $\|\theta\|_2 \leq \delta$ there exists $\theta_i \in \mathcal{N}_\epsilon$ with $\|\theta - \theta_i\|_2 \leq \epsilon$. Standard bounds [13, Lemma 5.7]) give

$$\log N(\epsilon, \delta\mathbb{B}_2^d, \|\cdot\|_2) \leq d \log \left(1 + \frac{2\delta}{\epsilon} \right). \quad (18)$$

For simplicity of notation and to avoid certain tedious negations, we define the “flipped”

monotone function family

$$\mathcal{F}_{\text{flip}} := \{g : \mathbb{R} \rightarrow \mathbb{R} \mid g(t) = \sigma(-t)\}_{\sigma \in \mathcal{F}_{\text{link}}}.$$

Now, for any $\theta \in \mathbb{R}^d$, let μ_θ denote the pushforward measure of $\langle V, \theta \rangle = Y \langle X, \theta \rangle$. For $\theta \in \mathbb{R}^d$, we then let $\mathcal{N}_{[\cdot], \gamma, \theta}$ be a minimal γ -bracketing of $\mathcal{F}_{\text{flip}}$ in the $L^4(\mu_\theta)$ norm. That is, for $N = N_{[\cdot]}(\gamma, \mathcal{F}_{\text{flip}}, L^2(\mu_\theta))$, we have $\mathcal{N}_{[\cdot], \gamma, \theta} = \{(l_{\theta, i}, u_{\theta, i})\}_{i=1}^N$, and for each $\sigma \in \mathcal{F}_{\text{link}}$, there exists $i = i(\sigma)$ such that

$$l_{\theta, i}(t) \leq \sigma(-t) \leq u_{\theta, i}(t) \quad \text{and} \quad \|u_{\theta, i} - l_{\theta, i}\|_{L^4(\mu_\theta)} \leq \gamma.$$

Because elements of $\mathcal{F}_{\text{flip}} \subset \mathbb{R} \rightarrow [0, 1]$ are monotone, van der Vaart and Wellner [12, Thm. 2.7.5] guarantee there exists a universal constant $K < \infty$ such that

$$\sup_Q \log N_{[\cdot]}(\gamma, \mathcal{F}_{\text{flip}}, L^4(Q)) \leq \frac{K}{\gamma}, \quad (19)$$

and in particular, $\log N_{[\cdot]}(\gamma, \mathcal{F}_{\text{flip}}, L^4(\mu_\theta)) \leq \frac{K}{\gamma}$ for each $\theta \in \mathbb{R}^d$.

With the covering $\mathcal{N}_\epsilon$ and induced bracketing collections $\mathcal{N}_{[\cdot], \gamma, \theta}$, we now turn to a construction of the actual bracketing of the class $\mathcal{F}_\delta^{(1)}$. For any $\theta \in \mathbb{R}^d$ and bracket $(l_{\theta, i}, u_{\theta, i}) \in \mathcal{N}_{[\cdot], \gamma, \theta}$, define the functionals $\widehat{l}_{\theta, j}, \widehat{u}_{\theta, j} : \mathbb{R}^d \rightarrow \mathbb{R}$ by

$$\widehat{l}_{\theta, j}(v) := (v_1)_+ \max \{l_{\theta, j}(\langle v, \theta \rangle) - \mathbb{L} \|v\|_2 \epsilon, 0\} - (-v_1)_+ \min \{u_{\theta, j}(\langle v, \theta \rangle) + \mathbb{L} \|v\|_2 \epsilon, 1\},$$

$$\widehat{u}_{\theta, j}(v) := (v_1)_+ \min \{u_{\theta, j}(\langle v, \theta \rangle) + \mathbb{L} \|v\|_2 \epsilon, 1\} - (-v_1)_+ \max \{l_{\theta, j}(\langle v, \theta \rangle) - \mathbb{L} \|v\|_2 \epsilon, 0\}.$$

The key is that these functions form a bracketing of $\mathcal{F}_\delta^{(1)}$:

Lemma I.6. *Define the set*

$$\mathcal{B}_{\epsilon, \gamma} := \left\{ \left(\widehat{l}_{\theta_i, j}, \widehat{u}_{\theta_i, j} \right) \mid \theta_i \in \mathcal{N}_\epsilon, \ 1 \leq j \leq N_{[\cdot]}(\gamma, \mathcal{F}_{\text{flip}}, L^4(\mu_{\theta_i})) \right\}.$$

Then $\mathcal{B}_{\epsilon, \gamma}$ is a

$$2\mathbb{L}\mathbb{E}[\|X\|_2^4]^{1/2} \cdot \epsilon + \mathbb{E}[\|X\|_2^4]^{1/4} \cdot \gamma$$

bracketing of $\mathcal{F}_\delta^{(1)}$ with cardinality at most $\log \text{card}(\mathcal{B}_{\epsilon,\gamma}) \leq \frac{K}{\gamma} + d \log(1 + \frac{\delta}{\epsilon})$.

Proof. Let $f_{\theta,\sigma}^{(1)}(v) \in \mathcal{F}_\delta^{(1)}$. Take $\theta_i \in \mathcal{N}_\epsilon$ satisfying $\|\theta - \theta_i\|_2 \leq \epsilon$ and $(l_{\theta_i,j}, u_{\theta_i,j}) \in \mathcal{N}_{[\cdot],\gamma,\theta_i}$ such that $l_{\theta_i,j}(t) \leq \sigma(-t) \leq u_{\theta_i,j}(t)$ for all t , where $\|u_{\theta_i,j} - l_{\theta_i,j}\|_{L^4(\mu_{\theta_i})} \leq \gamma$. We first demonstrate the bracketing guarantee

$$\widehat{l}_{\theta_i,j}(v) \leq f_{\theta,\sigma}^{(1)}(v) = v_1 \sigma(-\langle v, \theta \rangle) \leq \widehat{u}_{\theta_i,j}(v) \quad \text{for all } v \in \mathbb{R}^d.$$

For the upper bound, we have

$$\begin{aligned} f_{\theta,\sigma}^{(1)}(v) &= v_1 \sigma(-\langle v, \theta \rangle) \\ &\stackrel{(i)}{\leq} (v_1)_+ \min \{ \sigma(-\langle v, \theta_i \rangle) + \mathbf{L} |\langle v, \theta_i - \theta \rangle|, 1 \} - (-v_1)_+ \max \{ \sigma(-\langle v, \theta_i \rangle) - \mathbf{L} |\langle v, \theta_i - \theta \rangle|, 0 \} \\ &\stackrel{(ii)}{\leq} (v_1)_+ \min \{ \sigma(-\langle v, \theta_i \rangle) + \mathbf{L} \|v\|_2 \epsilon, 1 \} - (-v_1)_+ \max \{ \sigma(-\langle v, \theta_i \rangle) - \mathbf{L} \|v\|_2 \epsilon, 0 \} \\ &\stackrel{(iii)}{\leq} (v_1)_+ \min \{ u_{\theta_i,j}(\langle v, \theta_i \rangle) + \mathbf{L} \|v\|_2 \epsilon, 1 \} - (-v_1)_+ \max \{ l_{\theta_i,j}(\langle v, \theta_i \rangle) - \mathbf{L} \|v\|_2 \epsilon, 0 \} \\ &= \widehat{u}_{\theta_i,j}(v), \end{aligned}$$

where step (i) follows from the $\mathbf{L}$ -Lipschitz continuity of σ , (ii) from the Cauchy-Schwarz inequality and that $\|\theta - \theta_i\|_2 \leq \epsilon$, while step (iii) follows by the construction that $l_{\theta_i,j}(t) \leq \sigma(-t) \leq u_{\theta_i,j}(t)$ for all $t \in \mathbb{R}$. Similarly, we obtain the lower bound

$$f_{\theta,\sigma}^{(1)}(v) = v_1 \sigma(-\langle v, \theta \rangle) \geq \widehat{l}_{\theta_i,j}(v),$$

again valid for all $v \in \mathbb{R}^d$.

The second part of the proof is to bound the distance between the upper and lower elements in the bracketing. By definition, $\widehat{u}_{\theta_i,j} - \widehat{l}_{\theta_i,j}$ has the pointwise upper bound

$$\left(\widehat{u}_{\theta_i,j}(v) - \widehat{l}_{\theta_i,j}(v) \right)^2 \leq (|v_1| (u_{\theta_i,j}(\langle v, \theta_i \rangle) - l_{\theta_i,j}(\langle v, \theta_i \rangle)) + 2\mathbf{L} \|v\|_2 \epsilon)^2.$$

Recalling that $V = YX$, by the Minkowski and Cauchy-Schwarz inequalities, we thus obtain

$$\left\| \widehat{u}_{\theta_i,j}(V) - \widehat{l}_{\theta_i,j}(V) \right\|_{L^2(\mathbb{P})} \leq \| |V_1| (u_{\theta_i,j}(\langle V, \theta_i \rangle) - l_{\theta_i,j}(\langle V, \theta_i \rangle)) \|_{L^2(\mathbb{P})} + \| |V_1| \cdot 2\mathbf{L} \|V\|_2 \epsilon \|_{L^2(\mathbb{P})}$$

$$\leq \|V_1\|_{L^4(\mathbb{P})} \cdot \left(\|u_{\theta_i,j}(\langle V, \theta_i \rangle) - l_{\theta_i,j}(\langle V, \theta_i \rangle)\|_{L^4(\mathbb{P})} + 2\mathbb{L}\epsilon \cdot \|V\|_2 \|V\|_{L^4(\mathbb{P})} \right).$$

Noting the trivial bounds $\|V_1\|_{L^4(\mathbb{P})} \leq \|X\|_{L^4(\mathbb{P})} < \infty$ and the assumed bracketing distance

$$\|u_{\theta_i,j}(\langle V, \theta_i \rangle) - l_{\theta_i,j}(\langle V, \theta_i \rangle)\|_{L^4(\mathbb{P})} = \|u_{\theta_i,j} - l_{\theta_i,j}\|_{L^4(\mu_{\theta_i})} \leq \gamma,$$

we have the desired bracketing distance $\|\widehat{u}_{\theta_i,j} - \widehat{l}_{\theta_i,j}\|_{L^2(\mathbb{P})} \leq 2\mathbb{L}\mathbb{E}[\|X\|_2^4]^{1/2}\epsilon + \mathbb{E}[\|X\|_2^4]^{1/4}\gamma$.

The final cardinality bound is immediate via inequalities (18) and (19). $\square$

Lemma I.6 will yield the desired entropy integral bound. Fix any $t > 0$, and note that if we take $\epsilon = \epsilon(t) := t/(4\mathbb{L}\mathbb{E}[\|X\|_2^4]^{1/2})$ and $\gamma = \gamma(t) := t/(2\mathbb{E}[\|X\|_2^4]^{1/4})$, then the set $\mathcal{B}_{\epsilon,\gamma}$ is a t -bracketing of $\mathcal{F}_\delta^{(1)}$ in L^2 , and moreover, we have the cardinality bound

$$\log N_{[]} (t, \mathcal{F}_\delta^{(1)}, L^2(\mathbb{P})) \leq d \log \left(1 + \frac{2\delta}{\epsilon(t)} \right) + \frac{K}{\gamma(t)} \leq \frac{8\mathbb{L}d\delta \cdot \mathbb{E}[\|X\|_2^4]^{1/2} + 2K\mathbb{E}[\|X\|_2^4]^{1/4}}{t}.$$

Additionally, as covering numbers are necessarily integer, we have $\log N_{[]} (t) = 0$ whenever $t > (8\mathbb{L}d\delta \cdot \mathbb{E}[\|X\|_2^4]^{1/2} + 2K\mathbb{E}[\|X\|_2^4]^{1/4})/\log 2$. This gives the entropy integral bound

$$J_{[]} (\mathcal{F}_\delta^{(1)}, L^2(\mathbb{P})) = \int_0^\infty \sqrt{\log N_{[]} (t, \mathcal{F}_\delta^{(1)}, L^2(\mathbb{P}))} dt < \infty,$$

and consequently (cf. [11, Thm. 19.5] or [12, Ch. 2.5.2]), $\mathcal{F}_\delta^{(1)}$ is a Donsker class. A completely identical argument shows that $\mathcal{F}_\delta^{(i)}, i = 2, 3, \dots, d$ are Donsker, and so $\mathcal{F}_\delta$ is a Donsker class, completing the proof of the first claim in Lemma I.5.

To complete the proof of Lemma I.5, we need to show that

$$\mathbb{G}_{n,m}(\nabla_\theta \ell_{\widehat{\theta}_{n,m}^{\text{sp}}, \widehat{\sigma}_n} - \nabla_\theta \ell_{u^*, \sigma^*}) = \frac{1}{m} \sum_{j=1}^m \mathbb{G}_n^{(j)}(\nabla_\theta \ell_{\widehat{\theta}_{n,m}^{\text{sp}}, \sigma_{n,j}} - \nabla_\theta \ell_{u^*, \sigma_j^*}) \xrightarrow{p} 0,$$

where $\mathbb{G}_n^{(j)}$ denotes the empirical process on $(X_i, Y_{ij})_{i=1}^n$. Notably, because m is finite, it is sufficient to show that

$$\mathbb{G}_n^{(j)}(\nabla_\theta \ell_{\widehat{\theta}_{n,m}^{\text{sp}}, \sigma_{n,j}} - \nabla_\theta \ell_{u^*, \sigma_j^*}) \xrightarrow{p} 0, \quad j = 1, \dots, m.$$

To that end, we suppress dependence on j for notational simplicity and simply write $\mathbb{G}_n$ and $\ell_{\hat{\theta}_{n,m}^{\text{sp}}, \sigma_n}$, where $\|\sigma_n - \sigma^*\|_{L^2(\mathbb{P})} \xrightarrow{p} 0$. For any Donsker class $\mathcal{F} \subset \mathcal{X} \rightarrow \mathbb{R}^d$ and $\epsilon > 0$, we have

$$\limsup_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\|f-g\|_{L^2(\mathbb{P})} \leq \delta} \mathbb{G}_n(f-g) \geq \epsilon \right) = 0,$$

(see [3, Thm. 3.7.31]), and so in turn it is sufficient to prove that

$$\left\| \nabla \ell_{\hat{\theta}_{n,m}^{\text{sp}}, \sigma_n} - \nabla \ell_{u^*, \sigma^*} \right\|_{L^2(\mathbb{P})} \xrightarrow{p} 0. \quad (20)$$

To demonstrate the convergence (20), let M be finite, and note that for any fixed $\theta, \sigma \in \mathcal{F}_{\text{link}}$ that for $V = YX$ we have

$$\begin{aligned} \|\nabla \ell_{\theta, \sigma} - \nabla \ell_{u^*, \sigma^*}\|_{L^2(\mathbb{P})}^2 &= \mathbb{E} [\|V\sigma(-\langle V, \theta \rangle) - V\sigma^*(-\langle V, u^* \rangle)\|_2^2] \\ &\leq \mathbb{E} [\|V\|_2^2 1\{\|V\|_2 \geq M\}] + M^2 \|\sigma(-\langle V, \theta \rangle) - \sigma^*(-\langle V, u^* \rangle)\|_{L^2(\mathbb{P})}^2 \\ &\leq \mathbb{E} [\|X\|_2^2 1\{\|X\|_2 \geq M\}] + 2M^2 \|\sigma(-\langle V, u^* \rangle) - \sigma^*(-\langle V, u^* \rangle)\|_{L^2(\mathbb{P})}^2 \\ &\quad + 2M^2 \|\sigma(-\langle V, u^* \rangle) - \sigma(-\langle V, \theta \rangle)\|_{L^2(\mathbb{P})}^2 \end{aligned}$$

by the triangle inequality. As

$$\|\sigma(-\langle V, u^* \rangle) - \sigma(-\langle V, \theta \rangle)\|_{L^2(\mathbb{P})} \leq \mathbb{L} \|\theta - u^*\|_2 \cdot \|V\|_{L^2(\mathbb{P})} = \mathbb{L} \|\theta - u^*\|_2 \cdot \|X\|_{L^2(\mathbb{P})}$$

and $\hat{\theta}_{n,m}^{\text{sp}} - u^* \xrightarrow{p} 0$ and

$$\|\sigma_n(-\langle V, u^* \rangle) - \sigma^*(-\langle V, u^* \rangle)\|_{L^2(\mathbb{P})} = \|\sigma_n - \sigma^*\|_{L^2(\mathbb{P})} \xrightarrow{p} 0$$

by assumption, it follows that for any $\epsilon > 0$ that

$$\mathbb{P} \left(\left\| \nabla \ell_{\hat{\theta}_{n,m}^{\text{sp}}, \sigma_n} - \nabla \ell_{u^*, \sigma^*} \right\|_{L^2(\mathbb{P})} \geq \mathbb{E} [\|X\|_2^2 1\{\|X\|_2 \geq M\}] + \epsilon \right) \rightarrow 0.$$

Taking $M \uparrow \infty$ gives the convergence (20), completing the proof.

J Proofs for semiparametric approaches

J.1 Proof of Lemma H.1

As σ and σ^* are L -Lipschitz, we may without loss of generality assume that $L = 1$ and so $\|\sigma'_j\|_\infty \leq 1$ and $\|\sigma_j^{*\prime}\|_\infty \leq 1$. We can compute the Hessian at any $\theta \in \mathbb{R}^d$ and $\vec{\sigma} = (\sigma_1, \dots, \sigma_m)$,

$$\begin{aligned}\nabla^2 L(\theta, \vec{\sigma}) &= \mathbb{E} \left[\frac{1}{m} \sum_{j=1}^m (\sigma_j^*(-\langle X, u^* \rangle) \sigma'_j(\langle \theta, X \rangle) + \sigma_j^*(\langle X, u^* \rangle) \sigma'_j(-\langle \theta, X \rangle)) X X^\top \right] \\ &= \frac{1}{m} \sum_{j=1}^m \mathbb{E} [\sigma'_j(-Y \langle X, \theta \rangle) X X^\top],\end{aligned}$$

where in the last line we use that σ_j^* and σ_j are symmetric for $j = 1, \dots, m$. Therefore we can upper bound the distance between Hessians by

$$\|\nabla^2 L(\theta, \vec{\sigma}) - \nabla^2 L(u^*, \vec{\sigma}^*)\| \leq \frac{1}{m} \sum_{j=1}^m \underbrace{\|\mathbb{E} [\sigma'_j(-Y \langle X, \theta \rangle) X X^\top] - \mathbb{E} [\sigma_j^{*\prime}(-Y \langle X, u^* \rangle) X X^\top]\|}_{:=\delta_j},$$

and thus we only need to prove each quantity $\delta_j \rightarrow 0$ if $d_{\mathcal{F}_{\text{link}}^{\text{sp}}}((\theta, \vec{\sigma}), (u^*, \vec{\sigma}^*)) \rightarrow 0$, that is, if $\|\theta - u^*\|_2 \rightarrow 0$ and $\|\sigma_j(-Y \langle X, u^* \rangle) - \sigma_j^*(-Y \langle X, u^* \rangle)\|_{L^2(\mathbb{P})} \rightarrow 0$. In the following, we will show $\delta_j \rightarrow 0$ under the two different conditions. To further simplify the quantity, we claim it is sufficient to show $\xi_j := \|\mathbb{E}[\sigma'_j(-Y \langle X, \theta \rangle) X X^\top] - \mathbb{E}[\sigma_j^{*\prime}(-Y \langle X, u^* \rangle) X X^\top]\| \rightarrow 0$. Indeed, we have

Lemma J.1. *If $\xi_j \rightarrow 0$ and $\|\sigma_j(-Y \langle X, u^* \rangle) - \sigma_j^*(-Y \langle X, u^* \rangle)\|_{L^2(\mathbb{P})} \rightarrow 0$, then $\delta_j \rightarrow 0$.*

Proof. By the triangle inequality and the independent decomposition $X = Zu^* + W$, we have

$$\begin{aligned}\delta_j &\leq \xi_j + \|\mathbb{E} [\sigma'_j(-Y \langle X, u^* \rangle) X X^\top] - \mathbb{E} [\sigma_j^{*\prime}(-Y \langle X, u^* \rangle) X X^\top]\| \\ &= \xi_j + \left\| \mathbb{E} [(\sigma'_j(Z) - \sigma_j^{*\prime}(Z)) Z^2] \cdot u^* u^{*\top} \right\| = \xi_j + |\mathbb{E} [(\sigma'_j(Z) - \sigma_j^{*\prime}(Z)) Z^2]|.\end{aligned}$$

It remains to show $\mathbb{E} [(\sigma'_j(Z) - \sigma_j^{*\prime}(Z)) Z^2] \rightarrow 0$. Using the symmetry of σ_j and σ_j^* , so $\sigma'_j(t) = \sigma'_j(-t)$, we can replace Z by $|Z|$. Then integrating by parts, for any $0 < \epsilon < M < \infty$

we have

$$\begin{aligned}\mathbb{E} [\sigma'_j(Z)Z^2\mathbb{1}\{\epsilon \leq |Z| \leq M\}] &= \int_{\epsilon}^M \sigma'_j(z)z^2p(z)dz \\ &= \sigma_j(M)M^2p(M) - \sigma_j(\epsilon)\epsilon^2p(\epsilon) - \int_{\epsilon}^M \sigma_j(z)(2zp(z) + z^2p'(z))dz.\end{aligned}$$

By our w.l.o.g. assumption that $\|\sigma'\|_{\infty} \leq 1$, we have $|\mathbb{E}[\sigma'_j(Z)Z^2] - \mathbb{E}[\sigma'_j(Z)Z^2\mathbb{1}\{\epsilon \leq |Z| \leq M\}]| \leq \mathbb{E}[Z^2\mathbb{1}\{|Z| < \epsilon \text{ or } |Z| > M\}]$. Thus, recognizing the trivial bound $\|\sigma_j\|_{\infty} \leq 1$, we have

$$\begin{aligned}|\mathbb{E}[(\sigma'_j(Z) - \sigma_j^{\star'}(Z))Z^2]| &\leq 2\mathbb{E}[Z^2\mathbb{1}\{|Z| < \epsilon \text{ or } |Z| > M\}] + 2(\epsilon^2p(\epsilon) + M^2p(M)) + \\ &\quad + \underbrace{\left| \int_{\epsilon}^M \sigma_j(z)(2zp(z) + z^2p'(z)) - \int_{\epsilon}^M \sigma_j^{\star}(z)(2zp(z) + z^2p'(z)) \right|}_{(\star)}.\end{aligned}$$

We show for any fixed $0 < \epsilon < M < \infty$, $(\star) \rightarrow 0$. Applying the Cauchy-Schwarz inequality twice, we have the bounds

$$\begin{aligned}\left| \int_{\epsilon}^M (\sigma_j(z) - \sigma_j^{\star}(z))zp(z)dz \right| &\leq \|\sigma_j(Z) - \sigma_j^{\star}(Z)\|_{L^2(\mathbb{P})} \cdot \sqrt{\mathbb{E}[Z^2]} \rightarrow 0, \\ \left| \int_{\epsilon}^M (\sigma_j(z) - \sigma_j^{\star}(z))z^2p'(z)dz \right| &\leq \|\sigma_j(Z) - \sigma_j^{\star}(Z)\|_{L^2(\mathbb{P})} \cdot \sqrt{\int_{\epsilon}^M z^4 \left(\frac{p'(z)}{p(z)} \right)^2 p(z)dz} \\ &\leq \|\sigma_j(Z) - \sigma_j^{\star}(Z)\|_{L^2(\mathbb{P})} \cdot \sup_{z \in [\epsilon, M]} \left| \frac{p'(z)}{p(z)} \right| \sqrt{\mathbb{E}[Z^4]} \rightarrow 0,\end{aligned}$$

where for the final inequality we use that $p(z)$ is nonzero and continuously differentiable.

As $(\star) \rightarrow 0$, we evidently have $\limsup |\mathbb{E}[(\sigma'_j(Z) - \sigma_j^{\star'}(Z))Z^2]| \leq 2\mathbb{E}[Z^2(\mathbb{1}\{|Z| < \epsilon\} + \mathbb{1}\{|Z| > M\})] + 2(\epsilon^2p(\epsilon) + M^2p(M))$ for arbitrary $0 < \epsilon < M < \infty$. Using the assumptions that $\mathbb{E}[Z^2] \leq \mathbb{E}[\|X\|_2^2] < \infty$ and $\lim_{z \rightarrow s} z^2p(z) = 0$ for $s \in \{0, \infty\}$, we conclude the proof by taking $\epsilon \rightarrow 0$ and $M \rightarrow \infty$. $\square$

Finally we prove $\xi_j := \|\mathbb{E}[\sigma'_j(-Y\langle X, \theta \rangle)XX^{\top}] - \mathbb{E}[\sigma'_j(-Y\langle X, u^{\star} \rangle)XX^{\top}]\| \rightarrow 0$ under the two conditions in the statement of Lemma H.1: that σ'_j are Lipschitz or X has a continuous density.

Condition 1. The links have Lipschitz derivatives We apply Jensen's inequality to write

$$\begin{aligned}\xi_j &\leq \mathbb{E} [|\sigma'_j(-Y\langle X, \theta \rangle) - \sigma'_j(-Y\langle X, u^* \rangle)| \cdot \|XX^\top\|] = \mathbb{E} [|\sigma'_j(-Y\langle X, \theta \rangle) - \sigma'_j(-Y\langle X, u^* \rangle)| \cdot \|X\|_2^2] \\ &\leq \|\sigma'_j\|_{\text{Lip}} \|\theta - u^*\|_2 \cdot \mathbb{E} [\|X\|_2^3],\end{aligned}$$

where $\|\cdot\|_{\text{Lip}}$ denotes the Lipschitz constant of its argument. Taking $\theta \rightarrow u^*$ completes the proof for this case.

Condition 2. The covariates have continuous density Let X have density $q(x)$. We rewrite the convergence $\theta \rightarrow u^*$ instead as $\theta = Vu^*$ where $V \rightarrow I_d$ is invertible. We again divide the expectation into large $\|X\|_2$ part and small $\|X\|_2$ part. Let $M < \infty$ be large enough that $\mathbb{E}[\|X\|_2^2 \mathbf{1}\{\|X\|_2 > M\}] \leq \epsilon$. Then using $\|\sigma'_j\|_\infty < \infty$, we obtain

$$\begin{aligned}\xi_j &\leq \|\mathbb{E}[(\sigma'_j(-Y\langle X, \theta \rangle) - \sigma'_j(-Y\langle X, u^* \rangle))XX^\top] \mathbf{1}\{\|X\|_2 \leq M\}\| + 2L'\mathbb{E}[\|X\|_2^2 \mathbf{1}\{\|X\|_2 > M\}] \\ &\leq \|(V^{-\top}\mathbb{E}[\sigma'_j(-\langle V^\top X, u^* \rangle)V^\top XXV]V - \mathbb{E}[\sigma'_j(-\langle X, u^* \rangle)XX^\top]) \mathbf{1}\{\|X\|_2 \leq M\}\| + 2\|\sigma'_j\|_\infty \epsilon.\end{aligned}$$

By the linear transformation of variables $X' := V^\top X$, the first term in the above display is

$$\begin{aligned}&\|V^{-\top}\mathbb{E}[\sigma'_j(-Y\langle V^\top X, u^* \rangle)V^\top XXV]V - \mathbb{E}[\sigma'_j(-Y\langle X, u^* \rangle)XX^\top]\| \\ &= \left\| \det(V^{-1}) \cdot V^{-\top} \left(\int_{\|V^\top x\|_2 \leq M} \sigma'_j(-x^\top u^*)xx^\top p(V^{-\top}x)dx \right) V - \int_{\|x\|_2 \leq M} \sigma'_j(-x^\top u^*)xx^\top p(x)dx \right\|.\end{aligned}$$

Since $p(x)$ is absolutely continuous on any compact set, the above term converges to 0 as

$V \rightarrow I_d$. Finally, as $\epsilon > 0$ was arbitrary, we take $M \rightarrow \infty$ to conclude that $\xi_j \rightarrow 0$.

J.2 Proof of Proposition 10

We apply Theorem H.1. We first give the specialization of $C_{m,\vec{\sigma}^*}$ that the assumptions of the proposition imply, recognizing that as $\sigma^{\text{lr}'} = \sigma^{\text{lr}}(1 - \sigma^{\text{lr}})$, we have

$$C_{m,\vec{\sigma}^*} = \frac{\sum_{j=1}^m \mathbb{E}[\sigma^{\text{lr}}(\alpha_j^* Z)(1 - \sigma^{\text{lr}}(\alpha_j^* Z))]}{(\sum_{j=1}^m \mathbb{E}[\sigma^{\text{lr}'}(\alpha_j^* Z)])^2} = \left(\sum_{j=1}^m \mathbb{E}[\sigma^{\text{lr}}(\alpha_j^* Z)(1 - \sigma^{\text{lr}}(\alpha_j^* Z))] \right)^{-1}.$$

We now argue that we can actually invoke Theorem H.1, which requires verification of Assumption A1. Because for any $M < \infty$, the link functions $t \mapsto \sigma^{\text{lr}}(\alpha t)$ have Lipschitz continuous derivatives for $|\alpha| \leq M$, so when $\vec{\sigma}_n$ has form $\vec{\sigma}_n = [\sigma^{\text{lr}}(\alpha_{n,j} \cdot)]_{j=1}^m$, Lemma H.1 implies the continuity of the mapping $(\theta, \vec{\sigma}) \mapsto \nabla_{\theta}^2 L(\theta, \vec{\sigma})$ for $d_{\mathcal{F}_{\text{link}}^{\text{sp}}}$ at $(u^*, \vec{\sigma}^*)$. Then recognizing that by Lipschitz continuity of σ^{lr} we have

$$\|\vec{\sigma}_n(Z) - \vec{\sigma}^*(Z)\|_{L^2(\mathbb{P})} \leq \|\alpha_n - \alpha^*\|_2^2 \|Z\|_{L^2(\mathbb{P})}^2 \xrightarrow{p} 0$$

whenever $\alpha_n \in \mathbb{R}^m$ satisfies $\alpha_n \xrightarrow{p} \alpha^*$, we obtain the proposition.

K Proofs of nonparametric convergence results

In this technical appendix, we include proofs of the results from Section H.1 as well as a few additional results, which are essentially corollaries of results on localized complexities and nonparametric regression models, though we require a few modifications because our setting is slightly non-standard.

K.1 Preliminary results

To set notation and to make reading it self-contained, we provide some definitions. The $L^r(P)$ norm of a function or random vector is $\|f\|_{L^r(P)} = (\int |f|^r dP)^{1/r}$, so that its $L^2(P_n)$ -norm is $\|f\|_{L^2(P_n)}^2 = \frac{1}{n} \sum_{i=1}^n f(X_i)^2$. We consider the following abstract nonparametric regression

setting: we have a function class $\mathcal{F} \subset \{\mathbb{R} \rightarrow \mathbb{R}\}$ with $f^\star \in \mathcal{F}$, and our observations follow the model

$$Y_i = f^\star(X_i) + \xi_i, \quad (21)$$

but instead of observing (X_i, Y_i) pairs we observe $(\tilde{X}_i, Y_i)$ pairs, where $\tilde{X}_i$ may not be identical to X_i (these play the roll of $\langle u^\star, X_i \rangle$ versus $\langle u_n^{\text{init}}, X_i \rangle$ in the results to come). We assume that ξ_i are bounded so that $\sup \xi - \inf \xi \leq 1$, independent, and satisfy the conditional mean-zero property that $\mathbb{E}[\xi_i \mid X_i] = 0$ (though ξ_i may not be independent of X_i). For a (thus far unspecified) function class $\mathcal{F}$ we set

$$\hat{f} = \operatorname{argmin}_{f \in \mathcal{F}} P_n(Y - f(\tilde{X}))^2.$$

We now demonstrate that the error $\|\hat{f} - f^\star\|_{L^2(\mathbb{P}_n)}^2 = \frac{1}{n} \sum_{i=1}^n (\hat{f}(X_i) - f^\star(X_i))^2$ can be bounded by a combination of the errors $\tilde{X}_i - X_i$ and local complexities of the function class $\mathcal{F}$. Our starting point is a local complexity bound analagous to the localization results available for in-sample prediction error in nonparametric regression [cf. 13, Thm. 13.5]. To present the results, for a function class $\mathcal{H}$ we define the localized ξ -complexity

$$\mathcal{R}_n(u; \mathcal{H}) := \mathbb{E} \left[\sup_{h \in \mathcal{H}, \|h\|_{L^2(\mathbb{P}_n)} \leq u} \left| n^{-1} \sum_{i=1}^n \xi_i h(x_i) \right| \right],$$

where we treat the expectation conditionally on X_i and ξ_i are random (i.e. $\xi_i = Y_i - f^\star(X_i)$). For the model (21), we define the centered class $\mathcal{F}^\star = \{f - f^\star \mid f \in \mathcal{F}\}$, which is star-shaped as $\mathcal{F}$ is a convex set.¹ We say that δ satisfies the *critical radius inequality* if

$$\frac{1}{\delta} \mathcal{R}_n(\delta; \mathcal{F}^\star) \leq \delta. \quad (22)$$

With this, we can provide a proposition giving a high-probability bound on the in-sample prediction error of the empirical estimator $\hat{f}$, which is essentially identical to [13, Thm. 13.5],

¹A set $\mathcal{H}$ is *star-shaped* if for all $h \in \mathcal{H}$, if $\alpha \in [0, 1]$ then $\alpha h \in \mathcal{H}$.

though we require a few modifications to address that we observe $\tilde{X}_i$ and not X_i and that the noise ξ_i are bounded but not Gaussian.

Proposition 11. *Let $\mathcal{F}$ be a convex function class, $\delta_n > 0$ satisfy the critical inequality (22), and let $\gamma^2 = \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f(X_i) - f(\tilde{X}_i))^2$. Then for $t \geq \delta_n$,*

$$\mathbb{P} \left(\|\hat{f} - f^*\|_{L^2(\mathbb{P}_n)}^2 \geq 30t\delta_n + 25\gamma \max\{\gamma, \|\xi\|_{L^2(\mathbb{P}_n)}\} \mid X_1^n, \tilde{X}_1^n \right) \leq \exp \left(-\frac{nt\delta_n}{4} \right).$$

See Section K.3 for a proof of the proposition.

Revisiting the critical radius inequality (22), we can also apply [13, Corollary 13.7], which allows us to use an entropy integral to guarantee bounds on the critical radius. Here, we again fix $X_1^n = x_1^n$, and for a function class $\mathcal{H}$ we let $\mathcal{B}_n(\delta; \mathcal{H}) = \{h \in \text{Star}(\mathcal{H}) \mid \|h\|_{L^2(\mathbb{P}_n)} \leq \delta\}$. Let $N_n(t; \mathcal{B})$ denote the t -covering number of $\mathcal{B}$ in $\|n \cdot\|_{L^2(\mathbb{P}_n)}$ -norm. Then modifying a few numerical constants, we have

Corollary 1 (Wainwright [13], Corollary 13.7). *Let the conditions of Proposition 11 hold.*

Then for a numerical constant $C \leq 16$, any $\delta \in [0, 1]$ satisfying

$$\frac{C}{\sqrt{n}} \int_{\delta^2/4}^{\delta} \sqrt{\log N_n(t; \mathcal{B}_n(\delta, \mathcal{F}^*))} dt \leq \frac{\delta^2}{2}$$

satisfies the critical inequality (22).

As an immediate consequence of this inequality, we have the following:

Corollary 2. *Assume $|x_i| \leq b$ for all $i \in [n]$ and that $\mathcal{F}$ is contained in the class of M -Lipschitz functions with $f(0) = 0$ (or any other fixed constant). Then for a numerical constant $c < \infty$, the choice*

$$\delta_n = c \left(\frac{Mb}{n} \right)^{1/3}$$

satisfies the critical inequality (22).

Proof. The covering numbers N_∞ for the class $\mathcal{F}$ of M -Lipschitz functions on $[-b, b]$ satisfying $f(0) = 0$ in supremum norm $\|f\|_\infty = \sup_{x \in [-b, b]} |f(x)|$ satisfy $\log N_\infty(t; \mathcal{F}) \lesssim \frac{Mb}{t}$ [cf. 13, Example 5.10]. Using that $N_n \leq N_\infty$, we thus have

$$\int_{\delta^2/4}^{\delta} \sqrt{\log N_n(t; \mathcal{B}_n(\delta, \mathcal{F}^*))} \lesssim \int_{\delta^2/4}^{\delta} \sqrt{\frac{Mb}{t}} dt = 2\sqrt{Mb} \left(\sqrt{\delta} - \delta/4 \right) \leq 2\sqrt{Mb\delta}$$

whenever $\delta \leq 1$. For a numerical constant $c > 0$ it suffices in Corollary 1 to choose δ satisfying $c \frac{1}{\sqrt{n}} \sqrt{Mb\delta} \leq \delta^2$, or $\delta = c \left(\frac{Mb}{n} \right)^{1/3}$. $\square$

K.2 Proof of Proposition 9

We assume without loss of generality that $m = 1$, as nothing in the proof changes except notationally (as we assume m is fixed). We apply Proposition 11 and Corollary 2. For notational simplicity, let Q denote the measure on $\mathbb{R}$ that $Y \langle u^*, X \rangle$ induces for $X \sim P$, and Q_n similarly for P_n . We first show that $\|\sigma_n - \sigma_\star\|_{L^2(Q_n)}$ converges. First, we recall [6, Lemma 3] that $\max_{i \leq n} n^{-1/k} \|X_i\|_2 \xrightarrow{a.s.} 0$ as $n \rightarrow \infty$. Thus there is a (random) $B < \infty$ such that $\max_{i \leq n} \|X_i\|_2 \leq Bn^{1/k}$ for all n . Therefore, Corollary 2 implies that the choice $\delta_n = c \left(\frac{LB}{n^{1-1/k}} \right)^{1/3}$ satisfies the critical inequality (22), and taking $\gamma_n^2 = \frac{L^2}{n} \sum_{i=1}^n \langle u_n^{\text{init}} - u^*, X_i \rangle^2 \leq M^2 \epsilon_n^2 \|X\|_{L^2(\mathbb{P}_n)}^2$, we have that for $t \geq \delta_n$,

$$\|\sigma_n - \sigma_\star\|_{L^2(\mathbb{P}_n)}^2 \lesssim t\delta_n + L^2 \epsilon_n^2 \|X\|_{L^2(\mathbb{P}_n)}^2 \quad \text{with probability at least } 1 - \exp\left(-\frac{nt\delta_n}{4}\right)$$

on the event that $\max_{i \leq n} \|X_i\|_2 \leq Bn^{1/k}$ for all n , where we have conditioned (in the probability) on X_i . As $\|X\|_{L^2(\mathbb{P}_n)}^2 \xrightarrow{a.s.} \mathbb{E}[\|X\|_2^2]$, we may choose $t = \delta_n \gg 1/n^{1/3}$ and find that the Borell-Cantelli lemma then implies that with probability 1, there is a random $C < \infty$ such that

$$\|\sigma_n - \sigma_\star\|_{L^2(Q_n)}^2 \leq C \left(n^{\frac{2}{3k} - \frac{2}{3}} + \epsilon_n^2 \right) \quad (23)$$

for all n with probability 1.

Finally we argue that $\|\sigma_n - \sigma^\star\|_{L^2(Q)} \xrightarrow{a.s.} 0$. Let $b < \infty$ be otherwise arbitrary, and let $\mathcal{G}_b$ be the collection of $2L$ -Lipschitz functions on $[-b, b]$ with $\|g\|_\infty \leq 1$ and $g(0) = 0$ for $g \in \mathcal{G}_b$, noting that $\sigma_n - \sigma^\star$ restricted to $[-b, b]$ evidently belongs to $\mathcal{G}_b$. Then we have

$$\begin{aligned} & \|\sigma_n - \sigma^\star\|_{L^2(Q)}^2 \\ &= \|\sigma_n - \sigma^\star\|_{L^2(Q_n)}^2 + \int_{|t| \leq b} (\sigma_n(t) - \sigma^\star(t))^2 (dQ(t) - dQ_n(t)) + \int_{|t| > b} (\sigma_n(t) - \sigma^\star(t))^2 (dQ(t) - dQ_n(t)) \\ &\leq \|\sigma_n - \sigma^\star\|_{L^2(Q_n)}^2 + \sup_{g \in \mathcal{G}_b} |Qg^2 - Q_n g^2| + Q([-b, b]^c) + Q_n([-b, b]^c), \end{aligned} \quad (24)$$

where the inequality used that $\|\sigma_n - \sigma^\star\|_\infty \leq 1$ by construction. The first term in inequality (24) we have already controlled. We may control the second supremum term almost immediately using Dudley's entropy integral and a Rademacher contraction inequality. Indeed, we have

$$\mathbb{E}[\sup_{g \in \mathcal{G}_b} |Q_n g^2 - Q g^2|] \stackrel{(i)}{\leq} 2\mathbb{E}[\sup_{g \in \mathcal{G}_b} |Q_n^0 g^2|] \stackrel{(ii)}{\leq} 4\mathbb{E}[\sup_{g \in \mathcal{G}_b} |Q_n^0|] \stackrel{(iii)}{\lesssim} \frac{1}{\sqrt{n}} \int_0^\infty \sqrt{\log N_\infty(t; \mathcal{G}_b)} dt,$$

where inequality (i) is a standard symmetrization inequality, (ii) is the Rademacher contraction inequality [5, Ch. 4] applied to the function $t \mapsto t^2$, which is 2 Lipschitz for $t \in [-1, 1]$, and (iii) is Dudley's entropy integral bound. As the sup-norm log-covering numbers of L -Lipschitz functions on $[-b, b]$ scale as $\frac{Lb}{t}$ for $t \leq Lb$ and are 0 otherwise, we obtain $\mathbb{E}[\sup_{g \in \mathcal{G}_b} |Q_n g^2 - Q g^2|] \lesssim \frac{Lb}{\sqrt{n}}$. The bounded-differences inequality then implies that for any $t > 0$,

$$\mathbb{P}\left(\sup_{g \in \mathcal{G}_b} |(Q_n - Q)g^2| \geq c \frac{Lb}{\sqrt{n}} + t\right) \leq \mathbb{P}\left(\sup_{g \in \mathcal{G}_b} |(Q_n - Q)g^2| \geq \mathbb{E}[\sup_{g \in \mathcal{G}_b} |(Q_n - Q)g^2|] + t\right) \leq \exp(-cnt^2).$$

Finally, the final term in the bound (24) evidently satisfies $Q([-b, b]^c) \leq \frac{\mathbb{E}[\|X\|_2^b]}{b^k}$ and $\sup_b |Q_n([-b, b]^c) - Q([-b, b]^c)| \leq 2\sqrt{t/n}$ with probability at least $1 - e^{-2t^2}$ by the DKW inequality. We thus find by the Borel-Cantelli lemma that simultaneously for all $b < \infty$, with

probability at least $1 - e^{-nt^2}$ we have

$$\|\sigma_n - \sigma_\star\|_{L^2(Q)}^2 \leq \|\sigma_n - \sigma_\star\|_{L^2(Q_n)}^2 + \frac{CLb}{\sqrt{n}} + Ct + \frac{\mathbb{E}[\|X\|_2^k]}{b^k},$$

where C is a numerical constant.

Substituting inequality (23) into the preceding display and taking $b = n^{-\frac{1}{2(k+1)}}$, we get the result.

K.3 Proof of Proposition 11

We begin with an extension of the familiar basic inequality [e.g. 13, Eq. (13.18)], where we see by convexity that for any $\eta > 0$ we have

$$\begin{aligned} \sum_{i=1}^n (Y_i - \hat{f}(X_i))^2 &\leq (1 + \eta) \sum_{i=1}^n (Y_i - \hat{f}(\tilde{X}_i))^2 + (1 + 1/\eta) \sum_{i=1}^n (\hat{f}(\tilde{X}_i) - \hat{f}(X_i))^2 \\ &\leq (1 + \eta) \sum_{i=1}^n (Y_i - f^\star(\tilde{X}_i))^2 + (1 + 1/\eta) \sum_{i=1}^n (\hat{f}(\tilde{X}_i) - \hat{f}(X_i))^2 \\ &\leq (1 + \eta)^2 \sum_{i=1}^n (Y_i - f^\star(X_i))^2 + (2 + \eta + 1/\eta) \sum_{i=1}^n \left[(\hat{f}(\tilde{X}_i) - \hat{f}(X_i))^2 + (f^\star(\tilde{X}_i) - f^\star(X_i))^2 \right] \end{aligned}$$

Noting that $Y_i = f^\star(X_i) + \xi_i$, algebraic manipulations yield that if $\Delta = [f^\star(X_i) - \hat{f}(X_i)]_{i=1}^n$

is the error vector, then

$$\|\Delta\|_{L^2(\mathbb{P}_n)}^2 - \frac{2}{n} \xi^T \Delta + \|\xi\|_{L^2(\mathbb{P}_n)}^2 \leq (1 + \eta)^2 \|\xi\|_{L^2(\mathbb{P}_n)}^2 + \frac{2 + \eta + 1/\eta}{n} \sum_{i=1}^n \left[(\hat{f}(\tilde{X}_i) - \hat{f}(X_i))^2 + (f^\star(\tilde{X}_i) - f^\star(X_i))^2 \right]$$

Simplifying, we obtain the following:

Lemma K.1. *Let $\gamma^2 = \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f(X_i) - f(\tilde{X}_i))^2$ and $\Delta = [\hat{f}(X_i) - f^\star(X_i)]_{i=1}^n$. Then*

$$\begin{aligned} \|\Delta\|_{L^2(\mathbb{P}_n)}^2 &\leq \inf_{\eta} \left\{ (2\eta + \eta^2) \|\xi\|_{L^2(\mathbb{P}_n)}^2 + \frac{2}{n} \xi^T \Delta + (4 + 2\eta + 2/\eta) \gamma^2 \right\} \\ &\leq \frac{2}{n} \xi^T \Delta + 11\gamma \max \left\{ \gamma, \|\xi\|_{L^2(\mathbb{P}_n)} \right\}. \end{aligned}$$

Proof. The first inequality is algebraic manipulations and uses that our choice of η was arbitrary. For the second, we consider two cases: that $\|\xi\|_{L^2(\mathbb{P}_n)} \geq \gamma$ and that $\|\xi\|_{L^2(\mathbb{P}_n)} \leq \gamma$. In the first case, we consider $\eta \leq 1$, yielding the simplified bound that $\|\Delta\|_{L^2(\mathbb{P}_n)}^2 \leq \frac{2}{n}\xi^T \Delta + 3\eta \|\xi\|_{L^2(\mathbb{P}_n)}^2 + \frac{8\gamma^2}{\eta}$ for $\eta \leq 1$. Taking $\eta = \gamma / \|\xi\|_{L^2(\mathbb{P}_n)}$ gives that

$$\|\Delta\|_{L^2(\mathbb{P}_n)}^2 \leq \frac{2}{n}\xi^T \Delta + 11 \|\xi\|_{L^2(\mathbb{P}_n)} \gamma.$$

In the case that $\gamma \geq \|\xi\|_{L^2(\mathbb{P}_n)}$, we choose $\eta = 1$, and the bound simplifies to $\|\Delta\|_{L^2(\mathbb{P}_n)}^2 \leq \frac{2}{n}\xi^T \Delta + 3 \|\xi\|_{L^2(\mathbb{P}_n)}^2 + 8\gamma^2 \leq \frac{2}{n}\xi^T \Delta + 11\gamma^2$. $\square$

We now return to the proof of the proposition proper. We begin with an essentially immediate extension of the result [13, Lemma 13.12]. We let $\mathcal{H}$ be an arbitrary star-shaped function class. Define the event

$$A(u) := \left\{ \text{there exists } g \in \mathcal{H} \text{ s.t. } \|g\|_{L^2(\mathbb{P}_n)} \geq u \text{ and } |P_n \xi g| \geq 2 \|g\|_{L^2(\mathbb{P}_n)} u \right\}, \quad (25)$$

which we treat conditionally on $X_1^n = x_1^n$ as in the definition of the local complexity $\mathcal{R}_n$. (Here the noise ξ_i are still random, taken conditionally on $X_1^n = x_1^n$.)

Lemma K.2 (Modification of Lemma 13.12 of Wainwright [13]). *Let $\mathcal{H}$ be a star-shaped function class and let $\delta_n > 0$ satisfy the critical radius inequality*

$$\frac{1}{\delta} \mathcal{R}_n(\delta; \mathcal{H}) \leq \delta.$$

Then for all $u \geq \delta_n$, we have

$$\mathbb{P}(A(u)) \leq \exp\left(-\frac{nu^2}{4}\right).$$

Deferring the proof of Lemma K.2 to Section K.3.1, we can parallel the argument for [13, Thm. 13.5] to obtain our proposition.

Let $\mathcal{H} = \mathcal{F}^*$ in Lemma K.2. Whenever $t \geq \delta_n$, we have $\mathbb{P}(A(\sqrt{t\delta_n})) \leq e^{-nt\delta_n/4}$. We consider the two cases that $\|\Delta\|_{L^2(\mathbb{P}_n)} \leq \sqrt{t\delta_n}$. In the former that $\|\Delta\|_{L^2(\mathbb{P}_n)} \leq \sqrt{t\delta_n}$, we have nothing to do. In the latter, we have $\hat{f} - f^* \in \mathcal{F}^*$ while $\|\Delta\|_{L^2(\mathbb{P}_n)} > \sqrt{t\delta_n}$, so that if $A(\sqrt{t\delta_n})$ fails then we must have

$$\left| \frac{1}{n} \sum_{i=1}^n \xi_i \Delta(x_i) \right| \leq 2 \|\Delta\|_{L^2(\mathbb{P}_n)} \sqrt{t\delta_n}.$$

From the extension of the basic inequality in Lemma K.1 we see that

$$\|\Delta\|_{L^2(\mathbb{P}_n)}^2 \leq 4 \|\Delta\|_{L^2(\mathbb{P}_n)} \sqrt{t\delta_n} + 11 \max \left\{ \gamma^2, \gamma \|\xi\|_{L^2(\mathbb{P}_n)} \right\}.$$

Solving for $\|\Delta\|_{L^2(\mathbb{P}_n)}$ then yields

$$\|\Delta\|_{L^2(\mathbb{P}_n)} \leq \frac{4\sqrt{t\delta_n} + \sqrt{16t\delta_n + 44\gamma \max\{\gamma, \|\xi\|_{L^2(\mathbb{P}_n)}\}}}{2} \leq 4\sqrt{t\delta_n} + \sqrt{11\gamma \max\{\gamma, \|\xi\|_{L^2(\mathbb{P}_n)}\}}.$$

Simplifying gives Proposition 11.

K.3.1 Proof of Lemma K.2

Mimicking the proof of [13, Lemma 13.12], we begin with [13, Eq. (13.40)]:

$$\mathbb{P}(A(u)) \leq \mathbb{P}(Z_n(u) \geq 2u^2) \quad \text{for} \quad Z_n(u) := \sup_{g \in \mathcal{H}, \|g\|_{L^2(\mathbb{P}_n)} \leq u} \left| n^{-1} \sum_{i=1}^n \xi_i g(x_i) \right|.$$

Now note that if $\|g\|_{L^2(\mathbb{P}_n)} \leq u$, then the function $\xi \mapsto |n^{-1} \sum_{i=1}^n \xi_i g(x_i)|$ is $u/\sqrt{n}$ -Lipschitz with respect to the ℓ_2 -norm, so that convex concentration inequalities [e.g. 13, Theorem 3.4] imply that $\mathbb{P}(Z_n(u) \geq \mathbb{E}[Z_n(u)] + t) \leq \exp(-\frac{t^2 n}{4b^2 u^2})$ whenever $\sup \xi - \inf \xi \leq b$, and so for $b = 1$ we have

$$\mathbb{P}(Z_n(u) \geq \mathbb{E}[Z_n(u)] + u^2) \leq \exp\left(-\frac{nu^2}{4}\right).$$

As $\mathbb{E}[Z_n(u)] = \mathcal{R}_n(u)$, we finally use that the normalized complexity $t \mapsto \frac{\mathcal{R}_n(t)}{t}$ is non-decreasing [13, Lemma 13.6] to obtain that for $u \geq \delta_n$, $\frac{1}{u} \mathcal{R}_n(u) \leq \frac{1}{\delta_n} \mathcal{R}_n(\delta_n) \leq \delta_n$, the last

inequality by assumption. In particular, we find that for $u \geq \delta_n$ we have $\mathbb{E}[Z_n(u)] = \mathcal{R}_n(u) \leq u\delta_n \leq u^2$, and so

$$\mathbb{P}(Z_n(u) \geq 2u^2) \leq \mathbb{P}(Z_n(u) \geq \mathbb{E}[Z_n(u)] + u^2) \leq \exp\left(-\frac{nu^2}{4}\right),$$

as desired.

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